

Toward Provable Security in Anamorphic Extension: New Constructions and Analysis

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Abstract

In a dictatorial setting where the receiver's secret key may be exposed to an adversary, anamorphic encryption enables secure communication. Since its introduction in 2022, anamorphic cryptography has attracted considerable attention in the cryptographic literature. *Anamorphic extension* (AE) strengthens this paradigm by providing deniability: although the receiver can participate in covert communication, it can plausibly deny the existence of such communication to the dictator. The security of an AE of a *public key encryption* (PKE) is captured by the *indistinguishability between its normal and anamorphic modes of operation*, formalized by the IND-NA notion.

In this paper, we have introduced two concrete constructions of AE from the number-theoretic assumptions based indistinguishability against chosen plaintext attack (IND-CPA)-secure Goldwasser-Micali PKE and Benaloh PKE. We have proved the IND-NA security of our proposed AEs assuming the existence of secure *pseudo-random function* (PRF). To the best of our knowledge, our proposed Goldwasser-Micali based construction is the *first* AE that achieves natural robustness, attains bandwidth rate $\mathbf{1}$ and has small key-sizes where bandwidth rate is determined by the ratio of the covert and normal plaintext. The bandwidth rate of the Benaloh PKE-based construction is $\gg \mathbf{1}$. Our proposed AEs are efficient due to small key size, low computation costs mostly involving computations of PRF and modular operations, high bandwidth rate, low anamorphic ciphertext expansion rate (ratio of the ciphertext size and the covert plaintext size) and deniability.

Keywords: Pseudorandom function, Anamorphic extension, Goldwasser-Micali PKE, Benaloh PKE, IND-NA security

1 Introduction

The fundamental objective of cryptography has always been to guarantee secure communication between legitimate users over inherently insecure channels, despite the presence of potential eavesdroppers. In the contemporary, technologically driven world, however, a critical question frequently arises from governmental and regulatory bodies:

To what extent should individual privacy be permitted?

While absolute privacy ensures strong protection for users, it can also be exploited by malicious actors such as terrorists or adversarial entities to endanger public safety. Consequently, governments often reserve the authority to oversee or regulate the privacy of their citizens to mitigate such risks. Yet, this concentration of power may, in certain circumstances, foster authoritarian control, wherein governing authorities can compel users to disclose their secret keys, thereby gaining unrestricted access to all private communications. To preserve individual privacy under such potentially oppressive conditions, covert communication techniques - notably anamorphic encryption - have emerged as promising cryptographic mechanisms. These techniques enable secure and deniable information exchange, even in environments where privacy is subject to intrusive governmental oversight.

In 2022, Persiano et al. [1] introduced the notion of receiver *anamorphic public key encryption* (APKE), addressing scenarios in which the receiver's secret key may be compromised. Their construction operates in two distinct modes of operation, namely, (i) the normal mode and (ii) the anamorphic mode. In the normal mode of operation, the scheme functions as a standard PKE consisting of the triplet $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$. The receiver generates its secret and public key (sk, pk) running the algorithm PKE.Gen and publishes pk . The sender encrypts a plaintext m under pk using the encryption algorithm PKE.Enc , thereby obtaining a ciphertext ct which is transmitted to the receiver through an insecure communication channel. Upon receiving ct , the receiver runs the decryption algorithm PKE.Dec with sk to recover the original plaintext m . In addition to this standard mechanism, an anamorphic PKE triplet $\text{aPKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ is associated with the base PKE scheme to enable covert communication.

During anamorphic mode of operation, the receiver creates an additional double key dk along with an anamorphic secret-public key pair (ask, apk) and publishes apk . Any sender may encrypt a plaintext m under apk using the standard encryption algorithm PKE.Enc , producing a ciphertext ct that is computationally indistinguishable from a conventional ciphertext. The receiver can recover m with normal decryption algorithm PKE.Dec using ask .

Additionally, a sender may simultaneously encrypt a pair of normal and covert plaintext $(m, \hat{m})$ using the anamorphic encryption algorithm aPKE.Enc using dk , which implicitly contains apk . When the resulting ciphertext is decrypted using PKE.Dec and ask , only the normal plaintext m is obtained. In contrast, decryption via the anamorphic decryption algorithm aPKE.Dec with dk and ask reveals both the normal and the covert plaintexts.

Crucially, in a coercive environment where an adversarial authority (e.g., a dictatorial regime) forces a user to compromise its secret key, the user may safely disclose only the anamorphic secret key ask while keeping the double key dk hidden. This partial disclosure allows the adversary to recover the normal plaintext m , but prevents access to the covert plaintext $\hat{m}$, thereby preserving the confidentiality of hidden communications even under coercion.

The primary security requirements for an APKE are as follows: (i) the anamorphic secret-public key pair must be computationally indistinguishable from the normal secret-public key pair; (ii) the anamorphic ciphertext must be indistinguishable from a normal ciphertext; and (iii) the covert plaintext must not be recovered without access to the double key.

However, a critical limitation in the construction proposed by Persiano et al. [1] lies in its operational model. Specifically, to transition from the normal mode to the anamorphic mode, the receiver must update its secret-public key pair, an action that could appear suspicious to an adversarial authority or dictatorial regime monitoring key usage. To address this limitation, Banfi et al. [2] introduced the *anamorphic extension* (AE) of few existing PKE schemes. In their approach, the sender can generate a double key corresponding to any specific receiver by using only the public key of the receiver and the double key is securely transmitted to the receiver. Subsequently, the sender can choose to encrypt either a normal plaintext using the standard encryption algorithm or both the normal and covert plaintexts simultaneously using the anamorphic encryption algorithm. This extended scheme offers several additional features, enhancing both flexibility and deniability. Notably, the deniability of the double key is preserved as anyone can produce the double key using the receiver’s public key, enabling the receiver to deny ownership of a specific double key. Furthermore, the scheme allows the creation of multiple double keys corresponding to a single key pair, thereby increasing adaptability in diverse communication settings.

1.1 Literature Review

To provide privacy in spite of presence of a dictatorial authority who can access the receiver’s secret key, Persiano et al. [1] proposed the novel paradigm of APKE in the year of 2022. They introduced two generic constructions of receiver APKE. The first one relies on *rejection sampling* supporting very small plaintext while the second one is constructed using *Naor-Yung transform* [3] which uses an NIZK proof system. However, this approach suffers from the limitation that the double key can be generated a single time, only in conjunction with a newly generated public-secret key pair. Whenever the receiver wants a shift to anamorphic mode from normal mode, it needs to update its existing key-pair which is potentially suspicious to the dictator. To overcome this issue, Banfi et al. [2] in 2024, proposed the novel concept of AE of several existing PKEs enabling a user to correspond many double keys to a single existing key-pair of the receiver. The AE additionally features deniability where the receiver can plausibly deny ownership of a given double key as it is derived solely from the receiver’s public key. The concept of robustness was introduced in their work which ensures that the anamorphic decryption of normally encrypted plaintext returns decryption failure instead of returning a garbage covert plaintext unlike [1].

Furthermore, they have defined a new property called *selective randomness recoverability* (SRR) which ensures the existence of robust AE of a PKE. They construct AE of ElGamal [4], Cramer-Shoup [5] and OAEP [6] encryption using SRR property. In 2024, Catalano et al. [7] extended two generic and well-known encryption classes namely, hybrid encryption and the IBE-to-IND-CCA transform to support anamorphism. More importantly, they have proposed the notion of homomorphic anamorphic encryption where homomorphic properties hold for the covert plaintext as well as the normal one. They proposed APKE from homomorphic schemes including Cramer-Shoup lite PKE [5]. Additionally, they have proposed an anamorphic version of the fully-homomorphic PKE original GSW proposed by Gentry et al. [8]. Later in 2025, Catalano et al. [9] demonstrated that the rejection sampling-based APKE becomes insecure when instantiated over a (contrived, yet IND-CPA secure) PKE scheme. More broadly, they showed an impossibility result, showing that *stateless* black-box APKE *cannot* achieve full security where stateless refers to un-synchronized setting. Persiano et al. [10] in 2024, proposed an APKE from the IND-CCA secure *Koppula-Waters* (KW) scheme. They have further demonstrated how such an APKE naturally supports a hybrid anamorphic encapsulation mode. Dodis et al. [11] proposed the notion of *anamorphic-resistant* (AR) encryption, together with corresponding constructions and formal security definitions. They have shown that, within such schemes, any attempt to realize covert communication is either information-theoretically impossible or computationally infeasible. Interestingly, their work also formalizes three distinct models of dictatorial power - *Dictatoria*, *Warrantland* and *Privatopia* - capturing different levels of adversarial capabilities within the overarching dictatorial setting. Carnemolla et al. [12] provided concrete instantiations of AR encryption schemes and established that APKE is equivalent to algorithm-substitution attacks under suitable conditions. As a consequence, they reinterpreted their AR constructions as PKEs that are provably resistant to subversion attacks. Kutyłowski et al. [13] demonstrated that several widely deployed PKEs, including Goldwasser-Micali, Paillier and RSA-OAEP, admit randomness recovery and are therefore anamorphic under the general construction.

1.2 Our Contribution

In a dictatorial setting, the secrecy of keys and thus any meaningful notion of privacy is already compromised. Consequently, within such environments, the primary objective shifts toward enabling and securing covert communication. This paper aims to construct novel anamorphic versions based on existing well-known PKEs featuring higher bandwidth rate, low computational and storage cost, computational robustness and deniability of possession of the double key.

Most existing constructions of AE [2] suffer from a fundamental limitation: the covert plaintext is substantially *smaller* in size than the normal plaintext, thereby the *bandwidth rate* $\frac{|\hat{M}|}{|M|}$ is much lower where $|\hat{M}|$ and $|M|$ are the sizes of a covert plaintext $\hat{M}$ and a normal plaintext M , respectively. Furthermore, many of these schemes do not guarantee *computational robustness* that is the anamorphic decryption of a normal ciphertext in these schemes returns a meaningless covert plaintext instead of decryption failure.

We aim to address these challenges and present *two* AE of PKEs namely, Goldwasser-Micali cryptosystem [14] and Benaloh cryptosystem [15] based on number theoretic assumptions. Our proposed schemes are efficient as they exhibit low-computation cost, small public and secret key size and provide IND-NA security. We admit that due to comparatively higher ciphertext expansion rate, these PKEs have gained less practical importance in recent days. However, we emphasize that the large ciphertext expansion rate $\frac{|\text{ct}|}{|\text{M}|}$ in both of our constructions is utilized in a skillful way using *pseudo random function* (PRF) to support anamorphism where $|\text{ct}|$ and $|\text{M}|$ are the sizes of a ciphertext ct and a normal plaintext M , respectively. In other words, the inefficiencies are utilized to bring positive features. More interestingly, the covert plaintext size in both of our AE schemes is asymptotically comparable to the size of the normal plaintext, thereby improving the expressiveness of covert communication with increased bandwidth-rate. The double key sizes are extensively small in our schemes, unlike the existing ones [1, 2, 7, 10, 13]. Additionally, our first construction for AE inherently achieves natural computational robustness, ensuring reliable decryption without the need for additional mechanisms or modifications. To the best of our knowledge, this is the *first* computationally robust AE with bandwidth rate 1. Our second construction can be made computationally robust employing the techniques from Banfi et al. [2].

1.2.1 Anamorphic extension of Goldwasser Micali PKE

The Goldwasser Micali PKE [14] is a classical bit-wise PKE associated with plaintext space $\mathcal{M} = \{0, 1\}^n$ and the ciphertext space $\mathcal{C} = \mathbb{Z}_N^n$ and we propose the first computationally robust AE based on this, associated with the covert plaintext space $\widehat{\mathcal{M}} = \{0, 1\}^n$ and bandwidth rate 1. Here $\mathbb{Z}_N$ is the additive group of integers modulo N and $\mathbb{Z}_N^n$ is the collection of all n -tuples over $\mathbb{Z}_N$. The bit-wise structure of the Goldwasser-Micali PKE enables natural *computational robustness* in the proposed AE without any compromise in the covert plaintext size, offering the bandwidth rate $\frac{|\widehat{\text{M}}|}{|\text{M}|} = 1$ where $\text{M} \in \{0, 1\}^n$ and $\widehat{\text{M}} \in \{0, 1\}^n$ are the normal and covert plaintexts, respectively. The ciphertext expansion rate in the Goldwasser-Micali cryptosystem is large as $\frac{|\text{ct}|}{|\text{M}|} = \frac{n \log_2 N}{n} = \log_2 N$ where N is the modulus and n is the plaintext size of the underlying Goldwasser-Micali PKE. However, on a positive side, our anamorphic Goldwasser-Micali cryptosystem has low computation cost compared to existing similar schemes [1, 2, 7, 10, 13] as our anamorphic encryption requires n PRF-computation, $2n$ modular multiplication and anamorphic decryption requires computation of 1 Jacobian, 1 modular inverse, n PRF and n modular multiplication. Moreover, the double key size is $|\text{dk}| = 4 \log_2 N$ which is significantly less as compared to the existing ones [1, 2, 7, 10, 13].

1.2.2 Anamorphic extension of Benaloh cryptosystem

The Benaloh cryptosystem [15], associated with the plaintext space $\mathcal{M} = \mathbb{Z}_r$ and ciphertext space $\mathcal{C} = \mathbb{Z}_N^*$ where $N = pq$ for p, q being large primes such that $r \mid (p-1)$, $r \nmid (q-1)$, $r^2 \nmid (p-1)$ for a prime r , offers favorable computational efficiency with the

ciphertext expansion rate $\frac{|\text{ct}|}{|\text{M}|} = \frac{\log_2 N}{\log_2 r}$. We propose its anamorphic version associated with the covert plaintext space $\Delta = \{a^r \mid a \in \mathbb{Z}_N^*\}$ offering the anamorphic ciphertext expansion rate $\frac{|\text{ct}|}{|\text{M}|} = \frac{\log_2 N}{\log_2 N} = 1$, the bandwidth rate $\frac{|\hat{\text{M}}|}{|\text{M}|} = \frac{\log N}{\log r} \gg 1$ for $\text{M} \in \mathbb{Z}_r$ and $\hat{\text{M}} \in \Delta$ and the double key size $|\text{dk}| = 3 \log_2 N$ which add more efficiency than the existing works of [1, 2, 7, 10, 13]. This scheme has better computational efficiency than [1, 2, 7, 10, 13] since the anamorphic encryption needs 1 PRF computation, 2 modular multiplication and 1 modular exponentiation and the anamorphic decryption needs $r + 2$ modular exponentiation, 2 modular multiplication and 1 extended Euclidean algorithm computation.

Although we focus on two PKEs that are comparatively inefficient with respect to ciphertext size, we demonstrate how this apparent limitation can be leveraged into our advantage:

The inherent ciphertext expansion enables the seamless embedding of covert plaintext within the same ciphertext

Our anamorphic encryption schemes provide deniability as the sender can produce the double key by using the receiver's public key. Therefore, the receiver can participate in covert communication while retaining plausible deniability regarding ownership of the corresponding double key.

Additionally, our proposed AEs are synchronized where both the sender and the receiver hold a counter ctr and updates the same after every encryption and decryption, respectively. We can employ the technique proposed by Banfi et al. [2] to convert our synchronized schemes to unsynchronized versions. Since the covert plaintext space is a subspace of the encryption-randomness space in our second construction based on Benaloh PKE, it can be made robust following the work of Banfi et al. [2].

1.3 Technical Overview

Our central idea in proposing AE of existing PKEs is to subtly embed the covert plaintext within the encryption randomness. We provide a concise overview of our different techniques to incorporate the covert plaintext into the randomness. The core component in both our schemes is *secure PRF* with compatible input output spaces. A PRF $F : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$ is said to be *secure* if any *probabilistic polynomial time* (PPT) adversary can distinguish it from a uniformly random function $f : \mathcal{X} \rightarrow \mathcal{Y}$ only with negligible probability. We say that an AE of a PKE is IND-NA secure if its anamorphic mode is computationally indistinguishable from its standard mode of operation.

Though most of the existing anamorphic schemes use a PRF, they do not give any specification about how to access that PRF. In our AEs we keep the PRF description in an anamorphic public parameter which is generated once and for all with input the public key pk of the receiver in the underlying PKE. Besides using a PRF to incorporate the covert message in the encryption randomness, we have used the bit-wise encryption structure in a unique and skillful way to construct *robust AE* of the Goldwasser-Micali PKE for the first time. Similarly, to propose an AE for the Benaloh PKE, we have used the multiplicative group structure of the underlying group to

decrease computational cost and increase efficiency. We note that each PKE has its own elegant structure which may enable an easy and skillful anamorphic structure that demands the attention of the cryptographers.

1.3.1 Anamorphic Goldwasser-Micali cryptosystem

The standard Goldwasser-Micali cryptosystem $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is a bit-wise PKE. The algorithm $\Pi_{\text{GM}}.\text{Setup}(1^\lambda)$ returns the public parameter $\text{pp} = \mathcal{G}_\lambda$ for λ , being the security parameter where $\mathcal{G}_\lambda$ is λ -bit random prime number generator. The receiver possesses a pair of secret key $\text{sk} = (p, q)$ and public key $\text{pk} = (N, x)$ generated by $\Pi_{\text{GM}}.\text{Gen}(\text{pp})$ where p, q are primes generated by $\mathcal{G}_\lambda$, $N = pq$ and x is a *pseudo-quadratic residue* (PQR) modulo N for a security parameter λ . The sender encrypts a plaintext $M = m_1 \dots m_n \in \mathcal{M} = \{0, 1\}^n$ and produces a ciphertext $\text{ct} = c_1 \dots c_n \in \mathcal{C} = \mathbb{Z}_N^n$ by $\Pi_{\text{GM}}.\text{Enc}(\text{pp}, M, \text{pk})$ satisfying $c_i = y_i^2 x^{m_i}$ with randomly chosen $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ for all $i \in [n] = \{1, \dots, n\}$ ($\xleftarrow{\$}$ denotes uniformly random sampling).

In its AE $\Pi_{\text{aGM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ associated with the covert plaintext space $\widehat{\mathcal{M}} = \mathcal{M} = \{0, 1\}^n$, the sender (or the receiver) sets a PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ and keeps it in the anamorphic public parameter ap using the algorithm $\Pi_{\text{aGM}}.\text{Setup}(\text{pk})$ with input the public key $\text{pk} = (N, x)$ of the receiver. The anamorphic public parameter $\text{ap} = F$ is made public. The sender (or the receiver) runs the algorithm $\Pi_{\text{aGM}}.\text{Gen}(\text{ap}, \text{pk})$ with input the anamorphic public parameter $\text{ap} = F$ and the public key $\text{pk} = (N, x)$ of the receiver to generate an initial counter $\text{ctr} \in \mathbb{Z}_N$ and set the double key $\text{dk} = (K_1, K_2, \text{pk} = (N, x))$ by choosing randomly two distinct PRF keys $K_1, K_2 \in \mathbb{Z}_N$. The pair (dk, ctr) is communicated to the receiver (or the sender), through a secure communication channel. Note that to generate ap and dk , the public key pk of the receiver is used which gives the receiver the full potential to deny owning both ap and dk before the dictator.

To generate y_i , the sender uses the PRF key K_1 if the i -th bit of both the normal plaintext (m_i) and covert plaintext $(\widehat{m}_i)$ are the same. Otherwise, it uses K_2 . The generated $y_i = F(K_j, (\text{ctr}, m_i, i)) \in \mathbb{Z}_N^*$ for $j = 1$ or 2 , being the output of a PRF, acts like a random element from $\mathbb{Z}_N^*$ where $\text{ctr} \in \mathbb{Z}_N$ is the current counter. This ensures that the anamorphic ciphertext ct where

$$(\text{ct} = c_1 \dots c_n, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Enc}(\text{ap}, M, \widehat{M}, \text{dk}, \text{ctr})$$

with

$$c_i = y_i^2 x^{m_i} \bmod N, \quad y_i = F(K_j, (\text{ctr}, m_i, i)) \in \mathbb{Z}_N^*, \quad j = 1 \text{ or } 2, \quad i \in [n]$$

is *indistinguishable* from a normal ciphertext

$$\text{ct} = c_1 \dots c_n \leftarrow \Pi_{\text{GM}}.\text{Enc}(\text{pp}, M, \text{pk})$$

with

$$c_i = y_i^2 x^{m_i} \bmod N, \quad y_i \xleftarrow{\$} \mathbb{Z}_N^*, \quad i \in [n]$$

due to the security guaranty of the PRF F . Moreover, the $F(K_j, (\text{ctr}, m_i, i))$ can be calculated by the receiver by extracting K_j from dk and using the current counter $\text{ctr} \in \mathbb{Z}_N$ for $j = 1$ or 2 . The receiver does not know whether the i -th bit of both the normal and covert plaintexts are same or not. The receiver recovers the normal plaintext $\mathbf{M} = m_1 \dots m_n \in \{0, 1\}^n$ from the ciphertext using the normal decryption $\Pi_{\text{GM}}.\text{Dec}(\text{pp}, \text{ct}, \text{sk})$ of Goldwasser-Micali cryptosystem. To recover the covert plaintext $\hat{\mathbf{M}} = \hat{m}_1 \dots \hat{m}_n \in \{0, 1\}^n$, it calculates $z_i = c_i x^{-m_i} \bmod N$, extracting x from $\text{pk} = (N, x)$ contained in dk and checks whether z_i matches with $[F(K_1, (\text{ctr}, m_i, i))]^2 \bmod N$ or $[F(K_2, (\text{ctr}, m_i, i))]^2 \bmod N$ for each $i \in [n]$. If $z_i = [F(K_1, (\text{ctr}, m_i, i))]^2 \bmod N$ the receiver sets $\hat{m}_i = m_i$ and if $z_i = [F(K_2, (\text{ctr}, m_i, i))]^2 \bmod N$ the receiver sets $\hat{m}_i = 1 - m_i$. Otherwise, it sets $\hat{\mathbf{M}} = \perp$. Further, have proved the following security theorem.

Theorem 1.1 (Informal). *Our proposed AE $\Pi_{\text{aGM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ attains IND-NA security if a secure PRF exists.*

Robustness: If the sender normally encrypts a plaintext $\mathbf{M} = m_1 \dots m_n \in \{0, 1\}^n$, the ciphertext is $\text{ct} = c_1 \dots c_n$ where $c_i = y_i^2 x^{m_i}$ for $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ and the probability that $z_i = [F(K_j, (\text{ctr}, m_i, i))]^2 \bmod N$ for $j = 1$ or 2 is *negligible* $\forall i \in [n]$ from the security of the PRF F where $z_i = c_i x^{-m_i} \bmod N$. This ensures that anamorphic decryption of normal ciphertext ct returns $\perp$ with *overwhelming* probability and thus our proposed AE scheme Π_{aGM} achieves *computational robustness*.

1.3.2 Anamorphic Benaloh cryptosystem

The normal Benaloh cryptosystem $\Pi_{\text{Ben}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is a PKE with associated plaintext-space $\mathcal{M} = \mathbb{Z}_r$ where r is a prime number. The public parameter $\text{pp} = \mathcal{G}_\lambda$ is generated by the algorithm $\Pi_{\text{Ben}}.\text{Setup}(1^\lambda)$ where λ is the security parameter and $\mathcal{G}_\lambda$ generates λ -bit prime-pair (p, q) such that $r \mid (p-1), r^2 \nmid (p-1), r \nmid (q-1)$. The receiver owns a secret key $\text{sk} = (\phi, x)$ and public key $\text{pk} = (y, N)$ generated by the execution of $\Pi_{\text{Ben}}.\text{Gen}(\text{pp})$ where p, q are λ -bit primes and $N = pq$, $\phi = \Phi(N) = (p-1)(q-1)$, $y \xleftarrow{\$} \mathbb{Z}_N^*$ such that $y^{\phi/r} \not\equiv 1 \pmod{N}$ and $x = y^{\phi/r} \bmod N$. The sender invokes $\Pi_{\text{Ben}}.\text{Enc}(\text{pp}, \text{M}, \text{pk})$ to encrypt a plaintext $\text{M} \in \mathbb{Z}_r$ using the receiver's public key pk and produces the ciphertext $\text{ct} = y^{\text{M}} \hat{u} \bmod N$ where $\hat{u} = u^r \bmod N$ for randomly sampled $u \xleftarrow{\$} \mathbb{Z}_N^*$.

The AE $\Pi_{\text{aBen}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of Π_{Ben} is associated with the covert plaintext space $\hat{\mathcal{M}} = \Delta = \{a^r \mid a \in \mathbb{Z}_N^*\}$. The algorithm $\Pi_{\text{aBen}}.\text{Setup}(\text{pk})$ is run by the receiver (or the sender) on input of the receiver's public key $\text{pk} = (y, N)$ to generate and publish the anamorphic public parameters $\text{ap} = F$ containing a PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$. The algorithm $\Pi_{\text{aBen}}.\text{Gen}(\text{ap}, \text{pk})$ is run by either the receiver (or the sender) on input the anamorphic public parameter ap and the public key pk of the receiver to generate the double key $\text{dk} = (K, \text{pk})$ and the initial counter $\text{ctr} \in \mathbb{Z}_N$ which are communicated securely to the sender (or the receiver). Here $K \in \mathbb{Z}_N$ is a randomly chosen PRF key. Then the sender anamorphically

encrypts a normal plaintext $M \in \mathbb{Z}_r$ and a covert plaintext $\hat{M} \in \Delta$ using the algorithm $\Pi_{\text{aBen}}.\text{Enc}(\text{ap}, M, \hat{M}, \text{dk}, \text{ctr})$ with the receiver's anamorphic public parameter $\text{ap} = F$, double key $\text{dk} = (K, \text{pk})$ and the current counter ctr and produces an anamorphic ciphertext $\text{ct} = y^M \hat{u} \bmod N$ where $\hat{u} = \hat{M}F(K, (\text{ctr}, M)) \bmod N$. Since F is a secure PRF, $F(K, (\text{ctr}, M))$ is indistinguishable from a random element in Δ .

The original scheme Π_{Ben} sets $\hat{u} = u^r$ for randomly chosen $u \xleftarrow{\$} \mathbb{Z}_N^*$, making $\hat{u}$ a random element in Δ . In Π_{aBen} we set, $\hat{u} = \hat{M}F(K, (\text{ctr}, M)) \bmod N$ so that the covert plaintext $\hat{M}$ can be recovered from the randomness $\hat{u} \in \Delta$ by the owner of the double key $\text{dk} = (K, \text{pk})$. The PRF value $F(K, (\text{ctr}, M))$ brings the randomness into the component $\hat{u}$, F being a PRF. For decryption, the receiver first normally decrypts the ciphertext $\text{ct} \in \mathbb{Z}_N$ by running $\Pi_{\text{Ben}}.\text{Dec}(\text{pp}, \text{ct}, \text{sk})$ with its secret key sk and recovers the normal plaintext $M \in \mathbb{Z}_r$. The receiver recovers the randomness $\hat{u} = \text{ct}(y^{-1})^M \bmod N$ by extracting y from $\text{pk} = (y, N)$ contained in $\text{dk} = (K, \text{pk})$. Using the PRF F in the anamorphic public parameter ap and the current counter $\text{ctr} \in \mathbb{Z}_N$ the receiver computes the PRF value $F(K, (\text{ctr}, M))$ where the PRF key $K \in \mathbb{Z}_N$ is extracted from $\text{dk} = (K, \text{pk})$, applies the extended Euclidean algorithm to compute the inverse $[F(K, (\text{ctr}, M))]^{-1}$ and recovers the covert plaintext $\hat{M} = \hat{u}[F(K, (\text{ctr}, M))]^{-1}$. The correctness holds trivially. We have proved the following theorem.

Theorem 1.2 (Informal). *Our proposed AE $\Pi_{\text{aBen}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is IND-NA secure if a secure PRF exists.*

1.4 Comparison

Following the introduction of the primary concept of APKE in 2022, by Persiano et al. [1], AE was proposed by Banfi et al. [2] in 2024 to improve the mechanism for covert messaging in the dictatorial setting, making it a very active area of research in recent days for cryptographers.

Persiano et al. [1] proposed few APKE schemes, one of which is based on rejection sampling, offering the bandwidth rate $\frac{\log \lambda}{\lambda}$ that becomes significantly low for large security parameter λ . Their second construction with the bandwidth rate 1 relies on the Naor-Yung transform. They employed a generic simulation-sound NIZK proof system that is not cost-efficient. The double key in their second construction consists of a common reference string for the underlying NIZK along with two public keys and one secret key of a generic PKE and is remarkably large.

Banfi et al. [2] cleverly incorporated the covert plaintext into the encryption randomness and proposed the concept of *selective randomness recoverability* (SRR) and showed how every PKE with SRR supports AE with natural robustness. They instantiated several generic and concrete AEs of the ElGamal [4], Cramer-Shoup [5] and *optimal asymmetric encryption padding* (OAEP) encryption schemes where the constructions assume the *discrete logarithm problem* (DLP) to be computationally hard in the plaintext space $\mathcal{M}$ but efficiently solvable in the covert plaintext space $\hat{\mathcal{M}}$. As a consequence, the covert plaintext space size is required to be much smaller than the size of the normal plaintext space, making the bandwidth rate significantly lower. More

precisely, the normal plaintext space is exponentially larger than the covert plaintext space which significantly restricts the capacity of covert communication in these schemes. In the ElGamal-based AE scheme of [2], the double key $\mathbf{dk}$ contains one element of size $|\mathbb{H}|$ from the group $\mathbb{H}$, one generic PRF key K and a *lookup table* of size $l(|\mathbb{H}| + 1)$ which is not cost-efficient. The secret key, public key and the ciphertext are of size $|\mathbb{H}|$, $|\mathbb{H}|$ and $2|\mathbb{H}|$, respectively. The Cramer-Shoup-based AE of [2] has double key $\mathbf{dk}$ containing three group elements each of size $|\mathbb{H}|$, one generic PRF key K and a lookup table of size $l(|2\mathbb{H}| + 1)$. The secret key, public key and the ciphertext are of size $3|\mathbb{H}|$, $5|\mathbb{H}|$ and $4|\mathbb{H}|$, respectively. Here $l = |\widehat{\mathcal{M}}|$ is the covert plaintext space size, $\mathbb{H}$ is the underlying group for the scheme. As exhibited in Table 1, the large size of secret key, public key, double key and ciphertext increases the communication cost as compared to our proposed schemes. Since the double key needs to be communicated securely, its large size is a bottleneck of their schemes.

Catalano et al. [7] proposed two generic constructions of AE from two well-studied PKE paradigms namely, hybrid encryption and *identity-based encryption* (IBE)-to-*indistinguishable under chosen-ciphertext attacks* (IND-CCA) secure transform supporting robustness. They provided an instantiation of APKE with bandwidth rate $\frac{|\widehat{\mathcal{M}}|}{|\mathcal{M}|} = 1$ from the Cramer-Shoup lite PKE which requires 9 exponentiations in a large cyclic group $\mathbb{H}$ for encryption and a kind of rejection sampling for decryption. Therefore, this scheme incurs high computational cost. Both the normal and covert plaintext space sizes are comparatively *very small*. The secret key, public key, double key and ciphertext are of size $4|\mathbb{H}|$, $3|\mathbb{H}|$, $8|\mathbb{H}|$ and $4|\mathbb{H}|$, respectively. They proposed a second instantiation of APKE from the original GSW encryption scheme [8] under the *learning with error* (LWE) assumption. The public key, secret key, double key and ciphertext have size $mv \log_2 q$, $vs \log_2 q$, $vs \log_2 q$ and N^2 , respectively where m is the number of LWE samples, v is the lattice dimension and q is the modulus, $s = \lfloor \log_2 q \rfloor + 1$ and $U = vs$. This is computationally more costly compared to our constructions for AE since it involves 5 matrix multiplications and a kind of rejection sampling for encryption. Though the bandwidth rate is $\frac{|\widehat{\mathcal{M}}|}{|\mathcal{M}|} = 1$, the normal and covert plaintext spaces are very small in size. These two APKE constructions are *asymmetric* but does *not* support AE and thereby, lack deniability and flexibility.

Persiano et al. [10] have proposed an anamorphic version of the IND-CCA secure Koppula-Waters (KW) PKE with bandwidth rate $\frac{\lambda}{\text{poly}(\lambda)}$ using hinting PRF and an additional generic IND-CPA secure PKE where λ is a security parameter and $\text{poly}(\cdot)$ denotes polynomial function. The public key, secret key, double key and ciphertext are very large in size since they contain $O(\text{poly}(\lambda))$ number of elements.

Kutyłowski et al. [13] came up with three generic APKEs from the Goldwasser-Micali cryptosystem [14], ElGamal [4] and OAEP [6] encryption. All these constructions are instantiated using an additional generic IND-CPA-secure PKE, with the effective bandwidth rate determined entirely by the plaintext space of the underlying encryption scheme. Also, the cost efficiency relies on the efficiency of the underlying PKE scheme. Besides, these schemes are not AE and thereby, lack deniability of the double key. Their Goldwasser-Micali-based APKE exploits the recoverability of randomness from the ciphertext and achieves a bandwidth rate exceeding 1. In particular,

for each bit of the normal plaintext, the covert plaintext is drawn from the plaintext space of the underlying generic PKE. Consequently, the size of the covert plaintext must be at least as large as that of the normal plaintext.

Table 1 illustrates that our AE from Goldwasser-Micali PKE is secure assuming secure PRF but has bandwidth rate 1 which is less as compared to the existing Goldwasser-Micali PKE-based APKE [13]. However, our anamorphic cryptosystem is *more efficient* in terms of achieving *natural computational robustness* that is not attained by the previous work [1, 10, 13] does not require any generic cryptographic primitive unlike [1, 7, 10, 11, 13], has *less double key size* and *less computational cost* as it involves only modular multiplication, inversion, PRF and Jacobian computation (see Table 2). The public and secret keys have sizes $2 \log_2 N$ and $\log_2 N$, respectively, N being the integer modulus of the original Goldwasser-Micali PKE. Moreover, our double key has size $4 \log_2 N$ which is much smaller compared to [1, 2, 7, 10, 12, 13].

We have proposed another AE from Benaloh encryption with bandwidth rate $\gg 1$ based on secure PRF. Both the public and secret key have size $2 \log_2 N$ while the double key has size $3 \log_2 N$ which is significantly smaller than that of the existing ones [2, 7, 12] where N is the modulus of the original Benaloh PKE.

2 Preliminaries

Definition 1 (PKE) [2] A PKE $\Pi_{\text{PKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$, associated with plaintext space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ consisting of four PPT algorithms works as follows

$\text{pp} \leftarrow \text{Setup}(1^\lambda)$: The setup algorithm run by some trusted central body at the beginning to set the public parameters takes as input the security parameter λ and publishes the pp .

$(\text{sk}, \text{pk}) \leftarrow \text{Gen}(\text{pp})$: A user runs the key generation algorithm on input pp to generate its secret key sk and public key pk , publishes pk and keeps sk secret.

$\text{ct} \leftarrow \text{Enc}(\text{pp}, \text{M}, \text{pk})$: The sender runs the encryption algorithm with input the parameter pp , a plaintext $\text{M} \in \mathcal{M}$ and the public key pk of the receiver, generates a ciphertext $\text{ct} \in \mathcal{C}$ and sends it to the receiver. We refer ct generated by this algorithm as a normal ciphertext.

$\text{M}' \leftarrow \text{Dec}(\text{pp}, \text{ct}, \text{sk})$: The receiver executes the decryption algorithm on input the parameter pp , a ciphertext $\text{ct} \in \mathcal{C}$ and its own secret key sk and outputs a plaintext $\text{M}' \in \mathcal{M}$.

Remark 1 Note that in spite of using a number of public parameters, most of the existing PKE schemes do not explicitly mention the public parameters. We consider a separate setup algorithm to generate and fix those parameters which creates clarity for practical implementation.

Definition 2 (Correctness) [2] A PKE $\Pi_{\text{PKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is called correct if $\forall \lambda \in \mathbb{N}$ where $\mathbb{N}$ denotes the set of all natural numbers and all plaintext $\text{M} \in \mathcal{M}$,

$$\Pr \left[\text{M}' = \text{M} \mid \begin{array}{l} \text{pp} \leftarrow \Pi_{\text{PKE}}.\text{Setup}(1^\lambda) \\ (\text{sk}, \text{pk}) \leftarrow \Pi_{\text{PKE}}.\text{Gen}(\text{pp}) \\ \text{ct} \leftarrow \Pi_{\text{PKE}}.\text{Enc}(\text{pp}, \text{M}, \text{pk}) \\ \text{M}' \leftarrow \Pi_{\text{PKE}}.\text{Dec}(\text{pp}, \text{ct}, \text{sk}) \end{array} \right] = 1$$

Table 1 Comparative summary of IND-CPA secure anamorphic PKE schemes

Scheme	Type	pk	sk	dk	ct	$\alpha = \frac{ \tilde{M} }{ M }$	$\beta = \frac{ \text{ct} }{ M }$	$\gamma = \frac{ \text{ct} }{ M + \tilde{M} }$	Assumption
Π_{saE} [2]	AExt	$ \mathbb{H} $	$ \mathbb{H} $	$ K + l(\mathbb{H} + 1) + \mathbb{H} $	$2 \mathbb{H} $	$\frac{\log_2(\mathbb{H})}{ \mathbb{H} }$	2	$\frac{2 \mathbb{H} }{ \mathbb{H} + \log_2(\mathbb{H})}$	PRF, DLP
Π_{saCS} [2]	AExt	$3 \mathbb{H} $	$5 \mathbb{H} $	$ K + l(2 \mathbb{H} + 1) + 3 \mathbb{H} $	$4 \mathbb{H} $	$\frac{\log_2(\mathbb{H})}{ \mathbb{H} }$	4	$\frac{4 \mathbb{H} }{ \mathbb{H} + \log_2(\mathbb{H})}$	PRF, DLP
Π_{aCS} [7]	AEnc	$4 \mathbb{H} $	$3 \mathbb{H} $	$8 \mathbb{H} $	$4 \mathbb{H} $	$\frac{\log_2(\mathbb{H})}{ \mathbb{H} }$	4	$\frac{4 \mathbb{H} }{ \mathbb{H} + \log_2(\mathbb{H})}$	DLP
Π_{aGSW} [7]	AEnc	$mv \log_2 q$	$vs \log_2 q$	$vs \log_2 q$	U^2	1	mv	$\frac{mv}{2}$	LWE
Our Π_{aGM}	AExt	$2 \log_2 N$	$\log_2 N$	$4 \log_2 N$	$n \log_2 N$	1	$\log_2 N$	$\frac{\log_2 N}{2}$	QRP, PRF
Our Π_{aGen}	AExt	$2 \log_2 N$	$2 \log_2 N$	$3 \log_2 N$	$\log_2 N$	> 1	$\frac{\log_2 N}{\log_2 r}$	$\frac{\log_2 N}{\log_2 r + \log_2 N}$	rRP, PRF

AExt: anamorphic extension, AEnc: anamorphic encryption, $|\mathbb{H}|$: size of an element from the group $\mathbb{H}$, $|K|$: size of a generic key K in double key dk in [2], l : size of covert plaintext space in [2], q, v, m : parameters in original GSW encryption scheme where q is the modulus, v is the lattice dimension and m is the number of LWE samples, $s = \lfloor \log_2 q \rfloor + 1$, $U = vs$, N : modulus in Π_{aGM} and Π_{aGen} , n : plaintext size in Π_{aGM} , α : bandwidth rate, β : ciphertext expansion rate, γ : anamorphic ciphertext expansion rate, Π_{saE} : synchronized AE SyncAnamElGamal of ElGamal PKE in [2], Π_{aCS} : synchronized AE SyncAnamCramerShoup of Cramer Shoup PKE in [2], Π_{aCS} : APKE of Cramer Shoup PKE in [7], Π_{aGSW} : APKE of original GSW PKE in [7], PRF: pseudo-random function, DLP: discrete logarithm problem, LWE: learning with error problem, QRP: quadratic residuosity problem, rRP: r -th residuosity problem

Table 2 Comparative analysis of computational overheads in our AE schemes

Scheme	Enc	Dec
Our Π_{aGM}	$n(1 \text{PRF} + 2 \times)$	$1 J + 1 \text{Inv} + 1 \times + n(1 \text{PRF} + 1 \times)$
Our Π_{aBen}	$1 \text{PRF} + 2 \times + \text{Exp} $	$(r + 2) \text{Exp} + 2 \times + 2 \text{EE} $

n : plaintext size in Π_{aGM} , r : plaintext space size in Π_{aBen} , $|\text{PRF}|$: computation cost of PRF, $|\times|$: computation cost of multiplication, $|\text{Exp}|$: computation cost of exponentiation, $|\text{EE}|$: computation cost of extended Euclidean algorithm to find greatest common divisor of two integers or modular inverse of an integer, $|J|$: computation cost of Jacobian. The cost of random sampling, modular addition and bit-wise XOR operations are not taken into account due to very low computation cost

Definition 3 (IND-CPA security) [10] A PKE Π_{PKE} is called IND-CPA secure if for all PPT adversary $\mathcal{A}$ and any $\lambda \in \mathbb{N}$, a negligible function $\text{neg}(\cdot)$ exists such that

$$\text{Adv}_{\Pi_{PKE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) = \left| \Pr [\text{Exp}_{\Pi_{PKE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) = 1] - \frac{1}{2} \right| < \text{neg}(\lambda)$$

where the security experiment $\text{Exp}_{\Pi_{PKE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda)$ run between the adversary $\mathcal{A}$ and a challenger $\mathcal{C}$, is depicted in Fig. 1.

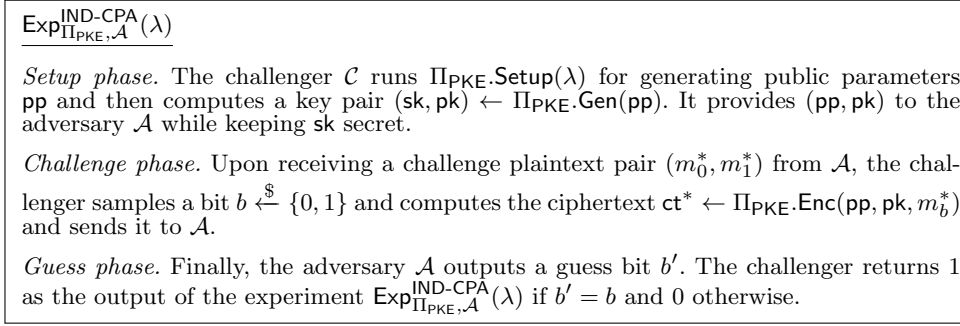


Fig. 1 $\text{Exp}_{\Pi_{PKE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda)$: Indistinguishability against CPA

Definition 4 (Negligible function) [10] A function $\mu : \mathbb{N} \rightarrow \mathbb{R}^+$ is called negligible if $\forall c \geq 0$, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $\mu(n) \leq n^{-c}$ where $\mathbb{R}^+$ is the set of all positive real numbers.

Definition 5 (AE (Synchronous)) [2] For a PKE $\Pi_{PKE} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ with corresponding plaintext space $\mathcal{M}$ and ciphertext space $\mathcal{C}$, its synchronous AE $\Pi_{aPKE} \equiv (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ associated with normal plaintext space $\mathcal{M}$, covert plaintext space $\widehat{\mathcal{M}}$ and ciphertext space $\mathcal{C}$, consisting of the following four additional PPT algorithms besides $\Pi_{PKE} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ works as follows

$\mathbf{ap} \leftarrow \text{Setup}(\mathbf{pk})$: The anamorphic setup algorithm run by a trusted authority takes as input the public key $\mathbf{pk}$ of the receiver generated by the algorithm $\Pi_{\text{PKE}}.\text{Gen}$ and publishes an anamorphic public parameter $\mathbf{ap}$. Here $(\mathbf{sk}, \mathbf{pk}) \leftarrow \Pi_{\text{PKE}}.\text{Gen}(\mathbf{pp})$ is the secret-public key pair of the receiver who has kept $\mathbf{sk}$ secret to itself and made $\mathbf{pk}$ publicly available and $\mathbf{pp} \leftarrow \Pi_{\text{PKE}}.\text{Setup}(1^\lambda)$ is the public parameter generated on input the security parameter λ at the beginning.

$(\mathbf{dk}, \text{ctr}) \leftarrow \text{Gen}(\mathbf{ap}, \mathbf{pk})$: The anamorphic key generation algorithm enables a user to generate a double key $\mathbf{dk}$ and a counter ctr on input the anamorphic parameter $\mathbf{ap}$ and a public key $\mathbf{pk}$ of a receiver. This algorithm can be run by the sender (or receiver) who sends the pair $(\mathbf{dk}, \text{ctr})$ to the receiver (or sender) securely.

$(\text{ct}, \text{ctr}) \leftarrow \text{Enc}(\mathbf{ap}, \mathbf{M}, \widehat{\mathbf{M}}, \mathbf{dk}, \text{ctr})$: The anamorphic encryption algorithm is executed by a sender who on input the anamorphic parameter $\mathbf{ap}$, a plaintext $\mathbf{M} \in \mathcal{M}$, a covert plaintext $\widehat{\mathbf{M}} \in \widehat{\mathcal{M}}$, a double key $\mathbf{dk}$ and a counter ctr , outputs a ciphertext $\text{ct} \in \mathcal{C}$ and an updated counter ctr . We refer ct generated by this algorithm as anamorphic ciphertext.

$(\widehat{\Gamma}/\perp, \text{ctr}) \leftarrow \text{Dec}(\mathbf{ap}, \text{ct}, \mathbf{sk}, \mathbf{dk}, \text{ctr})$: The decryption algorithm is run by the receiver who takes as input the anamorphic parameter $\mathbf{ap}$, a ciphertext $\text{ct} \in \mathcal{C}$, its secret key $\mathbf{sk}$, double key $\mathbf{dk}$ and a counter ctr and produces either $\widehat{\Gamma} \in \widehat{\mathcal{M}}$ or a special symbol $\perp$, indicating decryption failure and an updated counter ctr .

Note that the counter ctr is initialized once in the beginning and gets updated by the encryptor (resp. the decryptor) after each anamorphic encryption (resp. anamorphic decryption). More specifically at any point of time the encryptor and the decryptor hold the same counter. The anamorphic parameter and the double key are generated using only the public key of the receiver which ensures that the receiver has the full potential to deny the ownership of $\mathbf{ap}$ and $\mathbf{dk}$. Though the existing AE schemes do not specifically mention any anamorphic public parameter $\mathbf{ap}$, all of them use some common parameters. Therefore, in our definition we intend to formalize those parameters and generate through a proper setup algorithm.

Definition 6 (Correctness) [2] An AE $\Pi_{\text{aPKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of a PKE $\Pi_{\text{PKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is defined to be correct if $\forall \lambda \in \mathbb{N}$, all plaintext $\mathbf{M} \in \mathcal{M}$ and all covert plaintext $\widehat{\mathbf{M}} \in \widehat{\mathcal{M}}$,

$$\Pr \left[\widehat{\Gamma} = \widehat{\mathbf{M}} \mid \begin{array}{l} \mathbf{pp} \leftarrow \Pi_{\text{PKE}}.\text{Setup}(1^\lambda) \\ (\mathbf{sk}, \mathbf{pk}) \leftarrow \Pi_{\text{PKE}}.\text{Gen}(\mathbf{pp}) \\ \mathbf{ap} \leftarrow \Pi_{\text{aPKE}}.\text{Setup}(\mathbf{pk}) \\ (\mathbf{dk}, \text{ctr}) \leftarrow \Pi_{\text{aPKE}}.\text{Gen}(\mathbf{ap}, \mathbf{pk}) \\ (\text{ct}, \text{ctr}) \leftarrow \Pi_{\text{aPKE}}.\text{Enc}(\mathbf{ap}, \mathbf{M}, \widehat{\mathbf{M}}, \mathbf{dk}, \text{ctr}) \\ (\widehat{\Gamma}, \text{ctr}) \leftarrow \Pi_{\text{aPKE}}.\text{Dec}(\mathbf{ap}, \text{ct}, \mathbf{sk}, \mathbf{dk}, \text{ctr}) \end{array} \right] = 1$$

i.e., anamorphic ciphertext gets decrypted by anamorphic decryption algorithm with probability 1.

Definition 7 (Computational robustness) [2] An AE $\Pi_{\text{aPKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of a PKE $\Pi_{\text{PKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is defined to be computationally robust if $\forall \lambda \in \mathbb{N}$, a negligible function $\text{neg}(\cdot)$ exists so that for all plaintext $\mathbf{M} \in \mathcal{M}$,

$$\Pr \left[\widehat{\Gamma} \neq \perp \mid \begin{array}{l} \mathbf{pp} \leftarrow \Pi_{\text{PKE}}.\text{Setup}(1^\lambda) \\ (\mathbf{sk}, \mathbf{pk}) \leftarrow \Pi_{\text{PKE}}.\text{Gen}(\mathbf{pp}) \\ \mathbf{ap} \leftarrow \Pi_{\text{aPKE}}.\text{Setup}(\mathbf{pk}) \\ (\mathbf{dk}, \text{ctr}) \leftarrow \Pi_{\text{aPKE}}.\text{Gen}(\mathbf{ap}, \mathbf{pk}) \\ \text{ct} \leftarrow \Pi_{\text{PKE}}.\text{Enc}(\mathbf{pp}, \mathbf{M}, \mathbf{pk}) \\ (\widehat{\Gamma}, \text{ctr}) \leftarrow \Pi_{\text{aPKE}}.\text{Dec}(\mathbf{ap}, \text{ct}, \mathbf{sk}, \mathbf{dk}, \text{ctr}) \end{array} \right] < \text{neg}(\lambda)$$

i.e., a normal ciphertext can be anamorphically decrypted only with negligible probability.

Definition 8 (IND-NA security) [2] An AE $\Pi_{\text{aPKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of a PKE $\Pi_{\text{PKE}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is called IND-NA secure if $\forall \lambda \in \mathbb{N}$ and all PPT distinguisher $\mathcal{D}$, a negligible function $\text{neg}(\cdot)$ exists such that

$$\text{Adv}_{\Pi_{\text{aPKE}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) = \left| \Pr \left[\text{Exp}_{\Pi_{\text{aPKE}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) = 1 \right] - \frac{1}{2} \right| < \text{neg}(\lambda)$$

where the experiment $\text{Exp}_{\Pi_{\text{aPKE}}, \mathcal{D}}^{\text{IND-NA}}(\lambda)$ between the distinguisher $\mathcal{D}$ and a challenger $\mathcal{C}$ is presented in Fig. 2 for $b \in \{0, 1\}$.

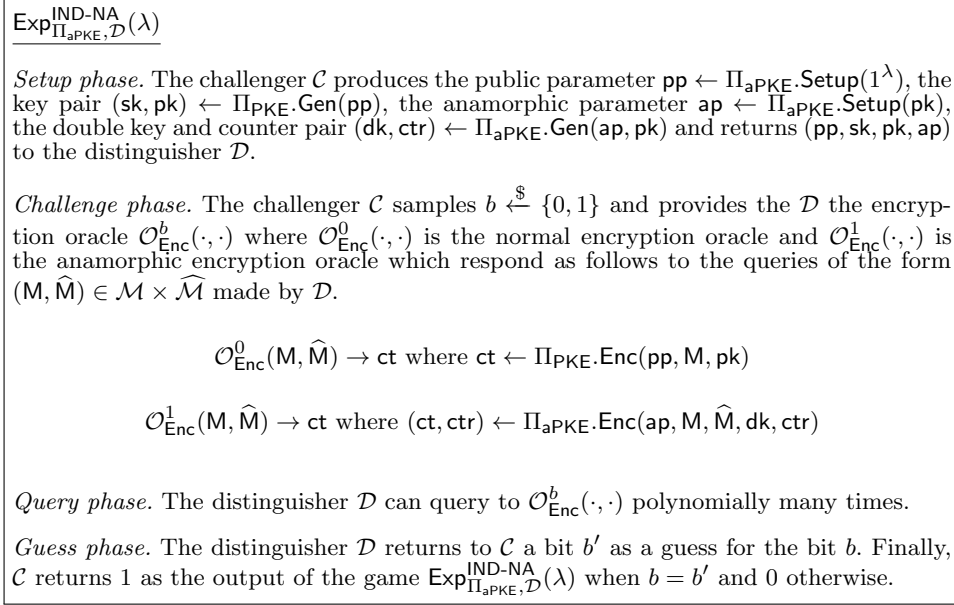


Fig. 2 $\text{Exp}_{\Pi_{\text{aPKE}}, \mathcal{D}}^{\text{IND-NA}}(\lambda)$: Indistinguishability between normal and anamorphic modes of encryption

Definition 9 (Quadratic residue (QR)) [14] An integer a with $\gcd(a, N) = 1$ where $N > 0$, is called a QR modulo N if $x^2 = a \bmod N$ for some integer x .

The set of all QRs modulo N is denoted by $\text{QR}(N)$.

Definition 10 (Legendre Symbol) [14] The Legendre symbol of an integer a modulo an odd prime p , denoted by $\left(\frac{a}{p}\right)$, is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \not\equiv 0 \pmod{p} \text{ and } \exists x \in \mathbb{Z} \text{ such that } x^2 \equiv a \pmod{p}, \\ -1 & \text{if } a \not\equiv 0 \pmod{p} \text{ and no such } x \text{ exists.} \end{cases}$$

Definition 11 (Jacobi Symbol) [14] Let $n > 1$ be an odd integer with prime factorization $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$ for p_i , being distinct primes and $e_i \geq 1$ for each i . The Jacobi symbol $\left(\frac{a}{n}\right)$ for an integer a , is defined as

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)^{e_1} \left(\frac{a}{p_2}\right)^{e_2} \dots \left(\frac{a}{p_k}\right)^{e_k}$$

where $\left(\frac{a}{p_i}\right)$ denotes the Legendre symbol. Additionally, for any integer a , we define $\left(\frac{a}{1}\right) = 1$. Consequently, for $n \geq 1$, the Jacobi symbol satisfies $\left(\frac{a}{n}\right) \in \{0, \pm 1\}$.

Definition 12 (PQR) [14] Let $N > 0$ and a are integers with $\gcd(a, N) = 1$. Then a is called a PQR modulo N if $\left(\frac{a}{N}\right) = 1$ and $\nexists x$ such that $x^2 = a \pmod{N}$.

The set of all PQRs modulo N is denoted by $\text{PQR}(N)$.

Definition 13 (Extended Euclidean algorithm) [16] This algorithm (Algorithm 1) computes the *greatest common divisor* (GCD) $\gcd(c, d)$ of two integers c, d and two integers u, v (called Bézout coefficients) such that:

$$\gcd(c, d) = uc + vd$$

Algorithm 1 ExtendedGCD(c, d) $\rightarrow$ ($\gcd, u, v$)

```

1: if ( $d == 0$ ) then
2:   return ( $c, 1, 0$ ) where  $c$  is the GCD and  $u = 1, v = 0$  are the Bézout coefficients
3: else
4:   ( $\gcd, u_1, v_1$ )  $\leftarrow$  ExtendedGCD( $d, c \bmod d$ );
5:    $u = v_1$ ;
6:    $v = u_1 - \lfloor c/d \rfloor \cdot v_1$ ;
7:   return ( $\gcd, u, v$ )
8: end if

```

Definition 14 (Secure PRF) [2] A PRF $F : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$ is said to be a secure if for every $K \in \mathcal{K}$, the function $F(K, \cdot)$ is indistinguishable from a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ chosen randomly from the collection $\mathcal{F}_{\mathcal{X}}^{\mathcal{Y}}$ of all functions from $\mathcal{X}$ to $\mathcal{Y}$. More specifically, $F : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$ is a secure PRF if $\forall \lambda \in \mathbb{N}$ and every PPT distinguisher $\mathcal{D}$, a negligible function $\text{neg}(\cdot)$ exists such that

$$\text{Adv}_{F, \mathcal{D}}^{\text{IND-PRF}}(\lambda) = \left| \Pr \left[\text{Exp}_{F, \mathcal{D}}^{\text{IND-PRF}}(\lambda) = 1 \right] - \frac{1}{2} \right| < \text{neg}(\lambda)$$

where the PRF security experiment $\text{Exp}_{F,\mathcal{D}}^{\text{IND-PRF}}$ between the distinguisher $\mathcal{D}$ and a challenger $\mathcal{C}$ is provided in Fig. 3 for $b \in \{0, 1\}$.

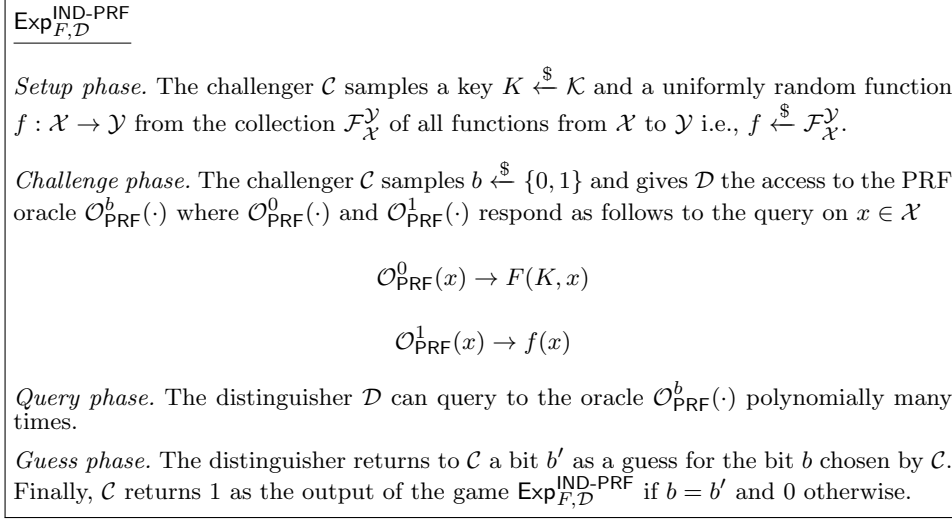


Fig. 3 $\text{Exp}_{F,\mathcal{D}}^{\text{IND-PRF}}$: PRF indistinguishability security experiment for PRF $F : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$

Definition 15 (Quadratic residuosity problem (QRP)) [14] Given a composite number $N = pq$ for unknown large primes p and q and an integer $a \in \mathbb{Z}_N^*$, the QRP is the problem to determine whether $a \in \text{QR}(N)$.

Definition 16 (r -th residuosity problem) [15] Given a composite number $N = pq$ for unknown large primes p and q , an integer $r \geq 2$ and an element $a \in \mathbb{Z}_N^*$, the r -th residuosity problem is to determine whether $\exists x \in \mathbb{Z}_N^*$ such that $x^r \equiv a \pmod{N}$. The integer r is called the residue degree.

This is a more generalized version of QRP.

2.1 Goldwasser-Micali cryptosystem

The Goldwasser-Micali PKE scheme $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of [14], associated with a plaintext-space $\mathcal{M} = \{0, 1\}^n$ and ciphertext-space $\mathcal{C} = \mathbb{Z}_N^n$ for an arbitrary positive integer n , consists of following four PPT algorithms.

- The algorithm $\Pi_{\text{GM}}.\text{Setup}$ presented in Algorithm 2 is run with input a security parameter λ by a trusted authority to publish the common public parameters pp which contains a prime number generator $\mathcal{G}_\lambda$ that generates λ -bit random primes.

Algorithm 2 $\Pi_{\text{GM}}.\text{Setup}(1^\lambda) \rightarrow \text{pp}$

Require: security parameter λ .

Ensure: public parameter pp .

- 1: Set a prime number generator $\mathcal{G}_\lambda$ that generates λ -bit primes;
 - 2: $\text{pp} = \mathcal{G}_\lambda$;
-

- A user runs the algorithm $\Pi_{\text{GM}}.\text{Gen}$ described in Algorithm 3 using the common public parameter $\text{pp} = \mathcal{G}_\lambda$ to generate its secret key $\text{sk} = (p, q)$ and the corresponding public key $\text{pk} = (N, x)$ where p, q are two λ -bit primes, $N = pq$ and $x \in \text{PQR}(N)$ is a PQR modulo N where

$$\text{PQR}(N) = \left\{ x \in \mathbb{Z}_N^* \mid \left(\frac{x}{p} \right) = \left(\frac{x}{q} \right) = -1, N = pq \right\}$$

and $\left(\frac{x}{p} \right), \left(\frac{x}{q} \right)$ denote the Legendre's symbols of x modulo p and modulo q , respectively (see Def. 10). The user keeps its secret key sk secret to itself and publishes the public key pk .

Algorithm 3 $\Pi_{\text{GM}}.\text{Gen}(\text{pp}) \rightarrow (\text{sk}, \text{pk})$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$.

Ensure: secret key sk and public key pk .

- 1: $p, q \xleftarrow{\$} \mathcal{G}_\lambda$ where p and q are two randomly chosen λ -bit primes;
 - 2: $N = pq$;
 - 3: $x \xleftarrow{\$} \text{PQR}(N)$ where $\text{PQR}(N)$ is the collection of all PQRs modulo N ;
 - 4: $\text{sk} = (p, q)$;
 - 5: $\text{pk} = (N, x)$;
-

- The encryptor runs the algorithm $\Pi_{\text{GM}}.\text{Enc}$ presented in Algorithm 4 on input of the public parameter $\text{pp} = \mathcal{G}_\lambda$, a plaintext $M \in \mathcal{M} = \{0, 1\}^n$ and the public key pk of the receiver, generates a ciphertext $\text{ct} = c_1 \dots c_n \in \mathcal{C} = \mathbb{Z}_N^n$ where $c_i = y_i^2 x^{m_i} \bmod N$ with $y_i \in \mathbb{Z}_N^*$ and sends it to the receiver through insecure channel.

Algorithm 4 $\Pi_{\text{GM}}.\text{Enc}(\text{pp}, M, \text{pk}) \rightarrow \text{ct}$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$, plaintext $M = m_1 \dots m_n \in \mathcal{M} = \{0, 1\}^n$ and public key $\text{pk} = (N, x)$.

Ensure: ciphertext $\text{ct} \in \mathbb{Z}_N^n$.

- 1: **for** ($i = 1$ to n) **do**
 - 2: $y_i \xleftarrow{\$} \mathbb{Z}_N^*$;
 - 3: $c_i = y_i^2 x^{m_i} \bmod N$;
 - 4: **end for**
 - 5: $\text{ct} = c_1 \dots c_n$;
-

- The decryptor runs the algorithm $\Pi_{\text{GM}}.\text{Dec}$ given in Algorithm 5 on input the public parameter $\text{pp} = \mathcal{G}_\lambda$, a ciphertext $\text{ct} \in \mathbb{Z}_N^n$ and its own secret key $\text{sk} = (p, q)$ and recovers the plaintext M' .

Algorithm 5 $\Pi_{\text{GM}}.\text{Dec}(\text{pp}, \text{ct}, \text{sk}) \rightarrow M'$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$, ciphertext $\text{ct} = c_1 \dots c_n \in \mathcal{C} = \mathbb{Z}_N^n$ and secret key $\text{sk} = (p, q)$.

Ensure: a plaintext M' .

```

1: for ( $i = 1$  to  $n$ ) do
2:   if  $\left(\left(\frac{c_i}{p}\right) = 1\right)$  then
3:      $m_i = 0$ ;
4:   else
5:      $m_i = 1$ ;
6:   end if
7: end for
8:  $M' = m_1 \dots m_n$ ;

```

Correctness. Note that for a single-bit plaintext $M = m \in \{0, 1\}$, the ciphertext is $\text{ct} = c \in \mathbb{Z}_N$ where $c = y^2 x^m \bmod N$ with $y \xleftarrow{\$} \mathbb{Z}_N^*$. As $y^2 \bmod N \in \text{QR}(N)$ and $x \in \text{PQR}(N)$, we have $\left(\frac{c}{N}\right) = 1$.

Case 1: ($M = m = 0$). The ciphertext in this case is $c = y^2 x^0 \bmod N = y^2 \bmod N \in \text{QR}(N)$ and $\left(\frac{c}{N}\right) = 1$ which implies

$$\left(\frac{c}{p}\right) = 1 \implies M' = 0 = M$$

(see lines 2 – 6 in Algorithm 5.)

Case 2: ($M = m = 1$). The ciphertext in this case is $c = y^2 x^1 \bmod N = y^2 x \bmod N \in \text{PQR}(N)$ since $x \in \text{PQR}(N)$ and $y^2 \bmod N \in \text{QR}(N)$. This implies

$$\left(\frac{c}{p}\right) = -1 \implies M' = 1 = M$$

(see lines 2 – 6 in Algorithm 5.)

Hence, the correctness of Π_{GM} holds for single-bit plaintext. The correctness for multi-bit plaintext is straight-forward.

Theorem 2.1 *The Goldwasser-Micali cryptosystem $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ achieves IND-CPA security (as per Def. 3) assuming the hardness of QRP (as per Def. 15).*

2.2 Benaloh cryptosystem

The Benaloh PKE $\Pi_{\text{Ben}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ of [15], associated with the plaintext-space $\mathcal{M} = \mathbb{Z}_r$ for a fixed prime number r and the ciphertext space $\mathcal{C} = \mathbb{Z}_N$ for $N = pq$

for two large primes p, q satisfying $r \mid (p-1)$, $r \nmid (q-1)$, $r^2 \nmid (p-1)$ consists of four PPT algorithms as described below.

- The algorithm $\Pi_{\text{Ben}}.\text{Setup}$ presented in Algorithm 6 is run once with input the security parameter λ in the beginning to set the public parameter $\text{pp} = \mathcal{G}_\lambda$ where $\mathcal{G}_\lambda$ is a prime-pair generator that generates pairs of large λ -bit random primes p, q satisfying $r \mid (p-1)$, $r \nmid (q-1)$, $r^2 \nmid (p-1)$. The parameter set pp is published publicly.

Algorithm 6 $\Pi_{\text{Ben}}.\text{Setup}(\lambda) \rightarrow \text{pp}$

Require: security parameter λ .

Ensure: public parameter pp .

- 1: Set a prime-pair generator $\mathcal{G}_\lambda$ that generates λ -bit prime-pair (p, q) such that $r \mid (p-1)$, $r^2 \nmid (p-1)$, $r \nmid (q-1)$ where r is a prime number and represents the cardinality of the plaintext space;
 - 2: $\text{pp} = \mathcal{G}_\lambda$;
-

- Each user generates its secret key $\text{sk} = (\phi, x)$ and public key $\text{pk} = (y, N)$ by running the algorithm $\Pi_{\text{Ben}}.\text{Gen}$ given in Algorithm 7 on input the public parameters $\text{pp} = \mathcal{G}_\lambda$ where (p, q) is the large prime-pair generated by $\mathcal{G}_\lambda$, $N = pq$, $\phi = (p-1)(q-1)$, y is a randomly chosen integer from $\mathbb{Z}_N^*$ such that $y^{\phi/r} \not\equiv 1 \pmod{N}$ and $x \equiv y^{\phi/r} \pmod{N}$. Then it keeps the secret key sk secret to itself and publishes the public key pk .

Algorithm 7 $\Pi_{\text{Ben}}.\text{Gen}(\text{pp}) \rightarrow (\text{sk}, \text{pk})$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$.

Ensure: secret key sk and public key pk .

- 1: $p, q \xleftarrow{\$} \mathcal{G}_\lambda$ where p and q are two randomly chosen λ -bit primes satisfying $r \mid (p-1)$, $r^2 \nmid (p-1)$, $r \nmid (q-1)$, r being the cardinality of the plaintext space;
 - 2: $N = pq$;
 - 3: $\phi = \Phi(N)$ where Φ is the Euler's totient function and $\Phi(N) = (p-1)(q-1)$;
 - 4: $y \xleftarrow{\$} \mathbb{Z}_N^*$ such that $y^{\phi/r} \not\equiv 1 \pmod{N}$;
 - 5: $x = y^{\phi/r} \pmod{N}$;
 - 6: $\text{sk} = (\phi, x)$;
 - 7: $\text{pk} = (y, N)$;
-

- The sender encrypts the plaintext $M \in \mathcal{M} = \mathbb{Z}_r$ for a fixed prime number r using the public parameters $\text{pp} = \mathcal{G}_\lambda$ and the receiver's public key pk by running the algorithm $\Pi_{\text{Ben}}.\text{Enc}$ described in Algorithm 8 and generates the ciphertext $\text{ct} \in \mathcal{C}$ and sends it to the receiver through public channel.

Algorithm 8 $\Pi_{\text{Ben}}.\text{Enc}(\text{pp}, \text{M}, \text{pk}) \rightarrow \text{ct}$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$, plaintext $\text{M} \in \mathcal{M} = \mathbb{Z}_r$ and public key $\text{pk} = (y, N)$.

Ensure: ciphertext ct .

- 1: $u \xleftarrow{\$} \mathbb{Z}_N^*$;
 - 2: $\text{ct} = y^{\text{M}} u^r \bmod N$;
-

- The receiver runs the algorithm $\Pi_{\text{Ben}}.\text{Dec}$ described in Algorithm 9 and decrypts the ciphertext $\text{ct} \in \mathcal{C} = \mathbb{Z}_N$ using the public parameters $\text{pp} = \mathcal{G}_\lambda$ and its own secret key sk and recovers the plaintext $\text{M}' \in \mathcal{M}$.

Algorithm 9 $\Pi_{\text{Ben}}.\text{Dec}(\text{pp}, \text{ct}, \text{sk}) \rightarrow \text{M}'$

Require: public parameter $\text{pp} = \mathcal{G}_\lambda$, ciphertext $\text{ct} \in \mathcal{C} = \mathbb{Z}_N$ and secret key $\text{sk} = (\phi, x)$.

Ensure: a plaintext M' .

- 1: $\text{M}' = \perp$;
 - 2: $a = \text{ct}^{\phi/r} \bmod N$;
 - 3: **for** ($i = 0$ to $r - 1$) **do**
 - 4: **if** ($x^i \equiv a \pmod{N}$) **then**
 - 5: $\text{M}' = i$;
 - 6: **break**;
 - 7: **end if**
 - 8: **end for**
-

Correctness. Note that the plaintext space is $\mathbb{Z}_r$ for some prime r and p, q are λ -bit prime numbers generated by $\mathcal{G}_\lambda$ for the security parameter λ satisfying $r \mid (p - 1)$, $r^2 \nmid (p - 1)$, $r \nmid (q - 1)$. The ciphertext $\text{ct} = y^{\text{M}} u^r \bmod N$ is generated using the public key $\text{pk} = (y, N)$ on a plaintext $\text{M} \in \mathbb{Z}_r$ by randomly sampling an integer $u \in \mathbb{Z}_N^*$ where $N = pq$ and $y \in \mathbb{Z}_N^*$ satisfying $y^{\phi/r} \neq 1 \bmod N$ with $\phi = (p - 1)(q - 1)$. The secret key is $\text{sk} = (\phi, x)$ with $x = y^{\phi/r} \bmod N$ and $u^\phi = 1 \bmod N$ by Euler's theorem as $u \in \mathbb{Z}_N^*$. Therefore,

$$a = \text{ct}^{\phi/r} \bmod N = (y^{\text{M}} u^r)^{\phi/r} \bmod N = y^{\frac{\text{M}\phi}{r}} u^\phi \bmod N = (y^{\phi/r})^{\text{M}} \bmod N = x^{\text{M}} \bmod N$$

(see line 2 in Algorithm 9). Also, $\text{M}' \in \mathbb{Z}_r$ such that $x^{\text{M}'} = a \bmod N$ (see lines 3 – 7 of Algorithm 9). Consequently,

$$x^{\text{M}'} = x^{\text{M}} \bmod N \implies x^{\text{M}' - \text{M}} = 1 \bmod N$$

Note that the order of x in the multiplicative group $\mathbb{Z}_N^*$ is r as r is a prime, $x = y^{\phi/r} \neq 1 \bmod N$ and $x^r = (y^{\phi/r})^r = y^\phi = 1 \bmod N$ by Euler's theorem.

Hence, $r \mid (\text{M}' - \text{M})$.

Note that $0 \leq \text{M}', \text{M} \leq r - 1$ since $\text{M}', \text{M} \in \mathbb{Z}_r$, yielding $\text{M}' - \text{M} = 0$ i.e., $\text{M}' = \text{M}$

$$\left[\begin{array}{l} \text{Thus, the correctness} \\ \text{Setup}(1^\lambda) \rightarrow \text{pp} = \mathcal{G}_\lambda \\ \text{Gen}(\text{pp}) \rightarrow (\text{sk} = (\phi, x), \text{pk} = (y, N)) \\ \text{Enc}(\text{pp}, M, \text{pk}) \rightarrow \text{ct} \\ \text{Dec}(\text{pp}, \text{ct}, \text{sk}) \rightarrow M' \end{array} \right] \text{ holds for the } \Pi_{\text{Ben}} =$$

Theorem 2.2 *The Benaloh cryptosystem $\Pi_{\text{Ben}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ as discussed in this Section, achieves IND-CPA security (as per Def. 3) assuming the hardness of the r -th residuosity problem (as per Def. 16).*

3 Our construction Π_{aGM} : synchronous AE from Goldwasser-Micali PKE

Consider the Goldwasser-Micali PKE Π_{GM} described in Section 2.1 associated with the plaintext space $\mathcal{M} = \{0, 1\}^n$ and the ciphertext space $\mathcal{C} = \mathbb{Z}_N^n$. We describe below its AE Π_{aGM} associated with the covert plaintext space $\widehat{\mathcal{M}} = \{0, 1\}^n$ that uses a collection of PRFs

$$\mathcal{F} = \{F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*\}$$

$$\text{Initially, } \Pi_{\text{GM}} = \left[\begin{array}{l} \text{Setup}(\lambda) \rightarrow \text{pp} \\ \text{Gen}(\text{pp}) \rightarrow (\text{sk}, \text{pk}) \\ \text{Enc}(\text{pp}, M, \text{pk}) \rightarrow \text{ct} \\ \text{Dec}(\text{pp}, \text{ct}, \text{sk}) \rightarrow M \end{array} \right] \text{ is in active mode where the normal pub-}$$

lic parameter $\text{pp} = \mathcal{G}_\lambda$ is generated by a trusted authority by using the algorithm $\Pi_{\text{GM}}.\text{Setup}$ where $\mathcal{G}_\lambda$ generates random λ -bit primes. A user runs the algorithm $\Pi_{\text{GM}}.\text{Gen}$ on input pp and generates its secret key $\text{sk} = (p, q)$ and the corresponding public key $\text{pk} = (N, x)$ where (p, q) is a pair of λ -bit primes generated by $\mathcal{G}_\lambda$, $N = pq$ and $x \in \text{PQR}(N)$ is a PQR modulo N . The secret key is kept secret to the user itself whereas the public key is published. The four PPT algorithms of the AE

$$\Pi_{\text{aGM}} = \left[\begin{array}{l} \text{Setup}(\text{pk}) \rightarrow \text{ap} \\ \text{Gen}(\text{ap}, \text{pk}) \rightarrow (\text{dk}, \text{ctr}) \\ \text{Enc}(\text{ap}, M, \widehat{M}, \text{dk}, \text{ctr}) \rightarrow (\text{ct}, \text{ctr}) \\ \text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \rightarrow (\widehat{M}, \text{ctr}) \end{array} \right] \text{ work as follows.}$$

- The algorithm $\Pi_{\text{aGM}}.\text{Setup}$ described in Algorithm 10 is run by the receiver with input the existing public key of a receiver $\text{pk} = (N, x)$ to set the anamorphic public parameters ap which contains a PRF $F \in \mathcal{F}$. The parameter set ap is published publicly.

Algorithm 10 $\Pi_{\text{aGM}}.\text{Setup}(\text{pk}) \rightarrow \text{ap}$

Require: public key $\text{pk} = (N, x)$.

Ensure: anamorphic public parameter ap .

- 1: $F \xleftarrow{\$} \mathcal{F}$ where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is an arbitrarily chosen PRF from $\mathcal{F}$;
 - 2: $\text{ap} = F$;
-

- The algorithm $\Pi_{\text{aGM}}.\text{Gen}$ given in Algorithm 11 is run by the sender (resp. the receiver) with input the anamorphic public parameter $\text{ap} = F$ and the public key $\text{pk} = (N, x)$ of the receiver (secret key holder) to generate a double key $\text{dk} = (K_1, K_2, \text{pk})$ and a counter ctr where $K_1, K_2 \in \mathbb{Z}_N$ and ctr is initialized with an integer modulo N . The double key dk and the counter ctr are communicated to the receiver (resp. the sender) through a secure channel.

Algorithm 11 $\Pi_{\text{aGM}}.\text{Gen}(\text{ap}, \text{pk}) \rightarrow (\text{dk}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$ and public key $\text{pk} = (N, x)$.

Ensure: double key dk and counter ctr .

- 1: $K_1, K_2 \xleftarrow{\$} \mathbb{Z}_N$;
 - 2: $\text{ctr} \xleftarrow{\$} \mathbb{Z}_N$;
 - 3: $\text{dk} = (K_1, K_2, \text{pk})$;
-

- The algorithm $\Pi_{\text{aGM}}.\text{Enc}$ presented in Algorithm 12 is run by the sender on input the anamorphic public parameters $\text{ap} = F$, a normal plaintext $M \in \mathcal{M}$, a covert plaintext $\hat{M} \in \hat{\mathcal{M}}$, the double key $\text{dk} = (K_1, K_2, \text{pk} = (N, x))$ of the receiver and a counter ctr and outputs a ciphertext $\text{ct} = c_1 \dots c_n$ and an updated counter $\text{ctr} = \text{ctr} + 1 \bmod N$ where $c_i = y_i^2 x^{m_i} \bmod N$ with $y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = m_i \\ F(K_2, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = 1 - m_i \end{cases}$. The ciphertext ct is sent through a public channel to the receiver whereas the updated counter ctr is kept secret to the sender.

Algorithm 12 $\Pi_{\text{aGM}}.\text{Enc}(\text{ap}, M, \hat{M}, \text{dk}, \text{ctr}) \rightarrow (\text{ct}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$, plaintext $M = m_1 \dots m_n \in \mathcal{M} = \{0, 1\}^n$, anamorphic plaintext $\hat{M} = \hat{m}_1 \dots \hat{m}_n \in \hat{\mathcal{M}} = \{0, 1\}^n$, double key $\text{dk} = (K_1, K_2, \text{pk} = (N, x))$ and current counter ctr .

Ensure: ciphertext $\text{ct} = c_1 \dots c_n \in \mathcal{C} = \mathbb{Z}_N^n$ and updated counter $\text{ctr} \in \mathbb{Z}_N$.

- 1: **for** ($i = 1$ to n) **do**
 - 2: **if** ($m_i = \hat{m}_i$) **then**
 - 3: $y_i = F(K_1, (\text{ctr}, m_i, i))$;
 - 4: **else**
 - 5: $y_i = F(K_2, (\text{ctr}, m_i, i))$;
 - 6: **end if**
 - 7: $c_i = y_i^2 x^{m_i} \bmod N$; $\triangleright y_i^2 \in \text{QR}(N)$ i.e., y_i^2 is a quadratic residue modulo N
 - 8: **end for**
 - 9: $\text{ct} = c_1 \dots c_n$;
 - 10: $\text{ctr} = \text{ctr} + 1 \bmod N$;
-

- The algorithm $\Pi_{\text{aGM}}.\text{Dec}$ given in Algorithm 13 is run by the receiver on input the anamorphic public parameters $\text{ap} = F$, the ciphertext $\text{ct} = c_1 \dots c_n \in \mathbb{Z}_N^n$, the secret key $\text{sk} = (p, q)$, the double key $\text{dk} = (K_1, K_2, \text{pk})$ and the current counter ctr to recover a covert plaintext $\hat{\Gamma} \in \hat{\mathcal{M}} \cup \{\perp\}$ and compute the updated counter ctr .

Algorithm 13 $\Pi_{\text{aGM}}.\text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \rightarrow (\hat{\Gamma}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$, ciphertext $\text{ct} = c_1 \dots c_n \in \mathcal{C} = \mathbb{Z}_N^n$, secret key $\text{sk} = (p, q)$, double key $\text{dk} = (K_1, K_2, \text{pk})$ and current counter ctr .

Ensure: covert plaintext $\hat{\Gamma} \in \hat{\mathcal{M}} \cup \{\perp\} = \{0, 1\}^n \cup \{\perp\}$ and updated counter ctr .

```

1: for  $(i = 1 \text{ to } n)$  do
2:   if  $\left(\left(\frac{c_i}{p}\right) = 1\right)$  then
3:      $m_i = 0$ ;
4:      $z_i = c_i$ ;
5:   else
6:      $m_i = 1$ ;
7:      $z_i = c_i x^{-1} \bmod N$ ;
8:   end if
9: end for
10:  $M = m_1 \dots m_n$ ;
11: for  $(i = 1 \text{ to } n)$  do
12:   if  $(z_i = (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N)$  then
13:      $\hat{\gamma}_i = m_i$ ;
14:   else
15:     if  $(z_i = (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N)$  then
16:        $\hat{\gamma}_i = \bar{m}_i = 1 - m_i$ ;
17:     else
18:        $\hat{\Gamma} = \perp$ ;
19:       break;
20:     end if
21:   end if
22: end for
23: if  $\hat{\Gamma} \neq \perp$  then
24:    $\hat{\Gamma} = \hat{\gamma}_1 \dots \hat{\gamma}_n$ ;
25: end if
26:  $\text{ctr} = \text{ctr} + 1 \bmod N$ ;

```

Theorem 3.1 *The AE $\Pi_{\text{aGM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ (presented above) of Goldwasser-Micali cryptosystem $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is computationally robust (as per Def. 7).*

Proof Let

$$\begin{aligned}
\text{pp} &= \mathcal{G}_\lambda \leftarrow \Pi_{\text{GM}}.\text{Setup}(1^\lambda) \\
(\text{sk} = (p, q), \text{pk} = (N, x)) &\leftarrow \Pi_{\text{GM}}.\text{Gen}(\text{pp}) \\
\text{ap} = F &\leftarrow \Pi_{\text{aGM}}.\text{Setup}(\text{pk}) \\
(\text{dk} = (K_1, K_2, \text{pk}), \text{ctr}) &\leftarrow \Pi_{\text{aGM}}.\text{Gen}(\text{ap}, \text{pk}) \\
\text{ct} = c_1 \dots c_n &\leftarrow \Pi_{\text{GM}}.\text{Enc}(\text{pp}, M, \text{pk})
\end{aligned}$$

where λ is security parameter, $\mathcal{G}_\lambda$ is a λ -bit prime generator, p, q are two λ -bit primes generated by $\mathcal{G}_\lambda$, $N = pq$, $x \in \text{PQR}(N)$ is a PQR modulo N , $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is a PRF from the PRF-family $\mathcal{F}$, $K_1, K_2 \in \mathbb{Z}_N$, $\text{ctr} \in \mathbb{Z}_N$ and $c_i = y_i^2 x^{m_i} \bmod N$ for uniformly random $y_i \in \mathbb{Z}_N^*$ for all $i \in [n]$.

Note that $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ in $\Pi_{\text{GM}}.\text{Enc}$ and $y_i = F(K_1, (\text{ctr}, m_i, i))$ or $y_i = F(K_2, (\text{ctr}, m_i, i))$ in $\Pi_{\text{aGM}}.\text{Enc} \forall i \in [n]$.

Now, $(\widehat{\Gamma}, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr})$ and $\widehat{\Gamma} \neq \perp$ when $z_i = (F(K_1, (\text{ctr}, m_i, i)))^2$ or $z_i = (F(K_2, (\text{ctr}, m_i, i)))^2$ (see lines 11 – 22 of Algorithm 13) where

$$z_i = \begin{cases} c_i & \text{if } m_i = 0 \\ c_i x^{-1} & \text{if } m_i = 1 \end{cases}$$

(see line 1 – 9 of Algorithm 13) $\forall i \in [n]$.

If $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ as in $\Pi_{\text{GM}}.\text{Enc}$ then

$$\begin{aligned} & \Pr \left[y_i^2 = (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N \vee y_i^2 = (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N \mid y_i \xleftarrow{\$} \mathbb{Z}_N^* \right] \\ &= \frac{2}{|\text{QR}(N)|} = \frac{2}{\Phi(N)/4} = \frac{8}{\Phi(N)} = \frac{8}{(p-1)(q-1)} \end{aligned}$$

As $m_i = 0$ means $c_i = y_i^2$ and $m_i = 1$ means $c_i = y_i^2 x$ (see lines 7 of Algorithm 12), we have

$$z_i = y_i^2 = \begin{cases} (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N & \text{if } \widehat{m}_i = m_i \\ (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N & \text{if } \widehat{m}_i = 1 - m_i \end{cases}$$

(see lines 2 – 6 of Algorithm 16) and consequently, we have

$$\begin{aligned} & \Pr \left[\widehat{\Gamma} \neq \perp \mid (\widehat{\Gamma}, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \right] \\ &= \prod_{i=1}^n \Pr \left[y_i^2 = (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N \vee y_i^2 = (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N \mid y_i \xleftarrow{\$} \mathbb{Z}_N^* \right] \\ &= \left(\frac{8}{(p-1)(q-1)} \right)^n \end{aligned}$$

As p, q are λ -bit primes each, we have

$$2^{\lambda-1} < p, q < 2^\lambda \implies \frac{1}{(2^\lambda - 1)^2} < \frac{1}{(p-1)(q-1)} < \frac{1}{(2^{\lambda-1} - 1)^2}$$

Therefore,

$$\Pr \left[\widehat{\Gamma} \neq \perp \mid (\widehat{\Gamma}, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \right] < \frac{8^n}{(2^{\lambda-1} - 1)^{2n}}$$

which is negligible in λ . Hence, our proposed AE Π_{aGM} of the existing Goldwasser-Micali cryptosystem Π_{GM} achieves computational robustness. $\square$

Theorem 3.2 *The AE $\Pi_{\text{aGM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ (presented above) of Goldwasser-Micali cryptosystem $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is correct (as per Def. 6).*

Proof Let

$$\begin{aligned} \text{pp} &= \mathcal{G}_\lambda \leftarrow \Pi_{\text{GM}}.\text{Setup}(1^\lambda) \\ (\text{sk} = (p, q), \text{pk} = (N, x)) &\leftarrow \Pi_{\text{GM}}.\text{Gen}(\text{pp}) \\ \text{ap} = F &\leftarrow \Pi_{\text{aGM}}.\text{Setup}(\text{pk}) \\ (\text{dk} = (K_1, K_2, \text{pk}), \text{ctr}) &\leftarrow \Pi_{\text{aGM}}.\text{Gen}(\text{ap}, \text{pk}) \end{aligned}$$

where λ is security parameter, $\mathcal{G}_\lambda$ is a λ -bit prime generator, p, q are two λ -bit primes generated by $\mathcal{G}_\lambda$, $N = pq$, $x \in \text{PQR}(N)$ is a PQR modulo N , $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is a random PRF chosen from the PRF-family $\mathcal{F}$, $K_1, K_2 \in \mathbb{Z}_N$ and $\text{ctr} \in \mathbb{Z}_N$.

Let

$$(\text{ct} = c_1 \dots c_n, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Enc}(\text{ap}, \mathbf{M}, \widehat{\mathbf{M}}, \text{dk}, \text{ctr})$$

for a covert plaintext $\widehat{\mathbf{M}} = \widehat{m}_1 \dots \widehat{m}_n \in \widehat{\mathcal{M}}$ and normal plaintext $\mathbf{M} = m_1 \dots m_n \in \mathcal{M}$ where $\forall i \in [n]$

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } m_i = \widehat{m}_i \\ F(K_2, (\text{ctr}, m_i, i)) & \text{if } m_i = 1 - \widehat{m}_i \end{cases}$$

Therefore,

$$c_i = \begin{cases} (F(K_1, (\text{ctr}, m_i, i)))^2 x^{m_i} \bmod N & \text{if } m_i = \widehat{m}_i \\ (F(K_2, (\text{ctr}, m_i, i)))^2 x^{m_i} \bmod N & \text{if } m_i = 1 - \widehat{m}_i \end{cases} \quad (1)$$

Let

$$(\widehat{\Gamma} = \widehat{\gamma}_1 \dots \widehat{\gamma}_n, \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Dec}(\text{ap}, \text{ct}, \text{dk}, \text{sk}, \text{ctr})$$

The normal plaintext $\mathbf{M} = m_1 \dots m_n$ is recovered by algorithm $\Pi_{\text{aGM}}.\text{Dec}$ (lines 1 – 10 of Algorithm 13) in the same way as in algorithm $\Pi_{\text{GM}}.\text{Dec}$ (lines 1 – 6 of Algorithm 5). The recovered normal plaintext $\mathbf{M}$ is correct by the correctness of Π_{GM} as shown in Section 2.1. Note that, for all $i \in [n]$

$$\widehat{\gamma}_i = \begin{cases} m_i & \text{if } c_i x^{-m_i} = (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N \\ \overline{m}_i & \text{if } c_i x^{-m_i} = (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N \end{cases} \quad (2)$$

(see lines 11 – 22 of Algorithm 13). Now, $\forall i \in [n]$,

$$\begin{aligned} \widehat{m}_i &= m_i \\ \implies c_i &= (F(K_1, (\text{ctr}, m_i, i)))^2 x^{m_i} \bmod N \text{ [from Eq. 1]} \\ \implies c_i x^{-m_i} &= (F(K_1, (\text{ctr}, m_i, i)))^2 \bmod N \\ \implies \widehat{\gamma}_i &= m_i = \widehat{m}_i \text{ [from Eq. 2]} \end{aligned}$$

and

$$\begin{aligned} \widehat{m}_i &\neq m_i \\ \implies \widehat{m}_i &= \overline{m}_i \\ \implies c_i &= (F(K_2, (\text{ctr}, m_i, i)))^2 x^{m_i} \bmod N \text{ [from Eq. 1]} \\ \implies c_i x^{-m_i} &= (F(K_2, (\text{ctr}, m_i, i)))^2 \bmod N \\ \implies \widehat{\gamma}_i &= \overline{m}_i = \widehat{m}_i \text{ [from Eq. 2]} \end{aligned}$$

Therefore, $\widehat{\Gamma} = \widehat{\gamma}_1 \dots \widehat{\gamma}_n = \widehat{m}_1 \dots \widehat{m}_n = \widehat{\mathbf{M}}$. Consequently, $\Pr[\widehat{\Gamma} = \widehat{\mathbf{M}}] = 1$ and the correctness of Π_{aGM} holds. $\square$

Theorem 3.3 *The AE $\Pi_{\text{aGM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ (presented above) of Goldwasser-Micali cryptosystem $\Pi_{\text{GM}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is IND-NA secure as per the security model described in Def. 8 assuming the existence of a secure PRF as per Def. 14.*

Proof We describe below the security game $\text{Game}_0 = \text{Exp}_{\text{aGM}, \mathcal{D}}^{\text{IND-NA}}$ between a challenger $\mathcal{C}$ and a distinguisher $\mathcal{D}$ following Fig. 2.

Setup phase: The challenger generates the public parameter $\text{pp} = \mathcal{G}_\lambda \leftarrow \Pi_{\text{GM}}.\text{Setup}(1^\lambda)$, a secret key and public key pair $(\text{sk} = (p, q), \text{pk} = (N, x)) \leftarrow \Pi_{\text{GM}}.\text{Gen}(\text{pp})$ and sends

$(\text{pp}, \text{sk}, \text{pk})$ to the distinguisher $\mathcal{D}$. Here $\mathcal{G}_\lambda$ is a prime number generator, p, q are λ -bit prime numbers generated by $\mathcal{G}_\lambda$, $N = pq$ and $x \in \text{PQR}(N)$ is a PQR modulo N .

The challenger $\mathcal{C}$ also generates the anamorphic public parameter $\text{ap} = F \leftarrow \Pi_{\text{aGM}}.\text{Setup}(\text{pk})$ and the double key-counter pair $(\text{dk} = (K_1, K_2, (N, x)), \text{ctr}) \leftarrow \Pi_{\text{aGM}}.\text{Gen}(\text{ap}, \text{pk})$ where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N$ is a randomly chosen PRF from the family of PRFs $\mathcal{F}$, $K_1, K_2 \in \mathbb{Z}_N$, $\text{ctr} \in \mathbb{Z}_N$ and ap is sent to $\mathcal{D}$. The challenger $\mathcal{C}$ samples $b \xleftarrow{\$} \{0, 1\}$.

Challenge phase: The challenger gives the distinguisher $\mathcal{D}$ access to the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ where

$$\mathcal{O}_{\text{Enc}}^b(\text{M}, \widehat{\text{M}}) \rightarrow \text{ct} = c_1 \dots c_n$$

with

$$\begin{aligned} \text{ct} &\leftarrow \Pi_{\text{GM}}.\text{Enc}(\text{pp}, \text{M}, \text{pk}) & \text{if } b = 0 \\ (\text{ct}, \text{ctr}) &\leftarrow \Pi_{\text{aGM}}.\text{Enc}(\text{ap}, \text{M}, \widehat{\text{M}}, \text{dk}, \text{ctr}) & \text{if } b = 1 \end{aligned}$$

for $(\text{M} = m_1 \dots m_n, \widehat{\text{M}} = \widehat{m}_1 \dots \widehat{m}_n) \in \mathcal{M} \times \widehat{\mathcal{M}} = \{0, 1\}^n \times \{0, 1\}^n$ and $\text{ctr} \in \mathbb{Z}_N$ is the current counter which stays unchanged and unutilized if $b = 0$ and gets updated after each oracle query to $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ if $b = 1$ as per Line 10 of Algorithm 12. Note that $\forall i \in [n]$, $c_i = y_i^2 x^{m_i} \bmod N$ with

$$y_i \xleftarrow{\$} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \widehat{m}_i = m_i \\ F(K_2, (\text{ctr}, m_i, i)) & \text{if } \widehat{m}_i = \bar{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

Query phase: The distinguisher $\mathcal{D}$ can query polynomially many times to the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ on pairs of normal and covert plaintexts $(\text{M}, \widehat{\text{M}}) \in \mathcal{M} \times \widehat{\mathcal{M}}$.

Guess phase: Finally, the distinguisher $\mathcal{D}$ outputs $b' \in \{0, 1\}$ as a guess for the bit b to challenger and wins the game if $b = b'$.

We define three more games **Game**₁, **Game**₂ and **Game**₃. Let E_i be the event that the distinguisher wins the game **Game** _{i} , $i = 0, 1, 2, 3$. The distinguisher $\mathcal{D}$ makes query of the form $(\text{M} = m_1 \dots m_n, \widehat{\text{M}} = \widehat{m}_1 \dots \widehat{m}_n) \in \mathcal{M} \times \widehat{\mathcal{M}} = \{0, 1\}^n \times \{0, 1\}^n$ to $\mathcal{C}$ in different games as shown below and receives different response ciphertext $\text{ct} = (c_1, \dots, c_n) \in \mathbb{Z}_N^n$ accordingly.

- **Game**₁: In this game, the distinguisher $\mathcal{D}$, on encryption oracle query $(\text{M} = m_1 \dots m_n, \widehat{\text{M}} = \widehat{m}_1 \dots \widehat{m}_n) \in \{0, 1\}^n \times \{0, 1\}^n$, receives the ciphertext $\text{ct} = (c_1, \dots, c_n)$ where

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i \xleftarrow{\$} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } m_i = \widehat{m}_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \bar{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

for all $i \in [n]$ and $f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ is uniformly random function, $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is a secure PRF and $K_1 \in \mathcal{K} = \mathbb{Z}_N$.

Claim 1 **Game**₁ is indistinguishable from **Game**₀.

Proof The games **Game**₀ and **Game**₁ are identical except that

$$y_i \stackrel{\$}{\leftarrow} \mathbb{Z}_N \quad \text{if } b = 0$$

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = m_i \\ F(K_2, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = \bar{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

in **Game**₀ whereas in **Game**₁,

$$y_i \stackrel{\$}{\leftarrow} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } m_i = \hat{m}_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \hat{m}_i = \bar{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

where $f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ is a uniformly random function and $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is a secure PRF. Thus,

$$y_i = \begin{cases} F(K_2, (\text{ctr}, m_i, i)) & \text{in Game}_0 \\ f_2(\text{ctr}, m_i, i) & \text{in Game}_1 \end{cases}$$

when $\hat{m}_i = 1 - m_i$ and $b = 1$.

Suppose the distinguisher $\bar{\mathcal{D}}$ distinguishes between **Game**₀ and **Game**₁ with probability $\frac{1}{2} + \delta'$ for some $\delta' \in \mathbb{R}$ such that $0 \leq \delta' \leq \frac{1}{2}$. We show below how to construct a PRF distinguisher $\mathcal{D}'$ for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ in the PRF security game $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$ (see Fig. 3) against PRF challenger $\mathcal{C}'$ invoking the distinguisher $\bar{\mathcal{D}}$ as a subroutine.

The PRF challenger $\mathcal{C}'$ samples $K_2 \stackrel{\$}{\leftarrow} \mathcal{K} = \mathbb{Z}_N$ and a uniformly random function $f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$. It samples $\hat{b} \stackrel{\$}{\leftarrow} \{0, 1\}$ and simulates the function oracle $\mathcal{O}_{\text{PRF}}^{\hat{b}}(\cdot)$ for the distinguisher $\mathcal{D}'$ as follows where $\mathcal{O}_{\text{PRF}}^0(\cdot)$ is the oracle for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ and $\mathcal{O}_{\text{PRF}}^1(\cdot)$ is the oracle for the uniformly random function $f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$.

$$\mathcal{O}_{\text{PRF}}^{\hat{b}}(\text{ctr}, m_i, i) = \begin{cases} F(K_2, (\text{ctr}, m_i, i)) & \text{if } \hat{b} = 0 \\ f_2(\text{ctr}, m_i, i) & \text{if } \hat{b} = 1 \end{cases}$$

Accordingly, $\mathcal{D}'$ samples $b \stackrel{\$}{\leftarrow} \{0, 1\}$ prepares the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ that on an encryption query on $(\mathbf{M} = m_1 \dots m_n, \hat{\mathbf{M}} = \hat{m}_1 \dots \hat{m}_n) \in \{0, 1\}^n \times \{0, 1\}^n$ from the distinguisher $\bar{\mathcal{D}}$ returns the ciphertext $\text{ct} = (c_1, \dots, c_n) \leftarrow \mathcal{O}_{\text{Enc}}^b(\mathbf{M}, \hat{\mathbf{M}})$ where

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i \stackrel{\$}{\leftarrow} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = m_i \\ \mathcal{O}_{\text{PRF}}^{\hat{b}}(\text{ctr}, m_i, i) & \text{if } \hat{m}_i = \bar{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

for $i \in [n]$ and $K_1 \in \mathcal{K} = \mathbb{Z}_N$ is a secure PRF key chosen by $\mathcal{D}'$. The distinguisher $\mathcal{D}'$ initializes the counter $\text{ctr} \stackrel{\$}{\leftarrow} \mathbb{Z}_N$ and increases the counter by 1 modulo N after each query if $b = 1$. The PRF distinguisher $\mathcal{D}'$ invokes $\bar{\mathcal{D}}$ to distinguish between **Game**₀ and **Game**₁. The query made by $\bar{\mathcal{D}}$ to the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ is simulated by the PRF distinguisher $\mathcal{D}'$ using the PRF oracle $\mathcal{O}_{\text{PRF}}^{\hat{b}}(\cdot)$ simulated by the PRF challenger $\mathcal{C}'$. Note the following cases.

$$\text{case } b = 1 \quad \hat{b} = 0 \text{ implies } y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = m_i \\ F(K_2, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = 1 - m_i \end{cases} \text{ as in Game}_0.$$

$$\hat{b} = 1 \text{ implies } y_i = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \hat{m}_i = m_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \hat{m}_i = 1 - m_i \end{cases} \text{ as in Game}_1.$$

case $b = 0$ $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ for $\hat{b} \in \{0, 1\}$ as in both Game_0 and Game_1 .

Thus, the encryption oracle $\mathcal{O}_{\text{Enc}}^b$ simulates Game_0 if $\hat{b} = 0$ and Game_1 if $\hat{b} = 1$.

Now, if the distinguisher $\mathcal{D}$ identifies the given game as Game_0 , $\mathcal{D}'$ outputs $\hat{b}' = 0$, else $\hat{b}' = 1$. Whenever $\mathcal{D}$ succeeds in distinguishing Game_0 and Game_1 , $\mathcal{D}'$ wins the experiment $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$. Therefore, $\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] \geq \frac{1}{2} + \delta'$ since we assumed that $\widehat{\mathcal{D}}$ distinguishes Game_0 and Game_1 with probability $\frac{1}{2} + \delta'$. Therefore,

$$\text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} = |\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] - \frac{1}{2}| \geq |\frac{1}{2} + \delta' - \frac{1}{2}| = \delta'$$

Since F is a secure PRF (Def. 14),

$$\delta' \leq \text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} \leq \epsilon_1(\lambda)$$

for some negligible function $\epsilon_1(\cdot)$. Thus any distinguisher $\widehat{\mathcal{D}}$ can distinguish between Game_0 and Game_1 only with probability negligibly close to $\frac{1}{2}$. Hence,

$$|\Pr[E_0] - \Pr[E_1]| < \epsilon_1(\lambda)$$

□

- **Game₂**: In this game, the distinguisher $\mathcal{D}$, on encryption oracle query ($M = m_1 \dots m_n, \widehat{M} = \widehat{m}_1 \dots \widehat{m}_n$) $\in \{0, 1\}^n \times \{0, 1\}^n$, receives the ciphertext $\text{ct} = (c_1, \dots, c_n)$ where

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i \xleftarrow{\$} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} f_1(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = m_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \overline{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

for all $i \in [n]$ and $f_1, f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ are two uniformly random functions.

Claim 2 Game_1 is indistinguishable from Game_2 .

Proof The Game_1 and Game_2 are identical except that for $b = 1$, $y_i = F(K_1, (\text{ctr}, m_i, i))$ in Game_1 whereas in Game_2 $y_i = f_1(\text{ctr}, m_i, i)$ when $m_i = \widehat{m}_i$ where $f_1 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ is a uniformly random function and $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ is a secure PRF. Suppose $\mathcal{D}$ distinguishes between Game_1 and Game_2 with probability $\frac{1}{2} + \delta$ for some $\delta \in \mathbb{R}$ such that $0 \leq \delta \leq \frac{1}{2}$ where $\mathbb{R}$ denotes the set of real numbers.

We construct below a PRF distinguisher $\mathcal{D}'$ for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ in the PRF security game $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$ (see Fig. 3) against PRF challenger $\mathcal{C}'$ and show how $\mathcal{D}'$ invokes the distinguisher $\mathcal{D}$.

Firstly, $\mathcal{C}'$ samples $K_1 \xleftarrow{\$} \mathcal{K} = \mathbb{Z}_N$ and a uniformly random function $f_1 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$. The PRF challenger $\mathcal{C}'$ samples $\widehat{b} \xleftarrow{\$} \{0, 1\}$, and gives $\mathcal{D}$ the access to the PRF oracle $\mathcal{O}_{\text{PRF}}^{\widehat{b}}(\cdot)$ where $\mathcal{O}_{\text{PRF}}^0(\cdot)$ is the oracle for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \{0, 1\} \times [n]) \rightarrow \mathbb{Z}_N^*$ and $\mathcal{O}_{\text{PRF}}^1(\cdot)$ is the oracle for the uniformly random function $f_1 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$. Thus,

$$\mathcal{O}_{\text{PRF}}^{\widehat{b}}(\text{ctr}, m_i, i) = \begin{cases} F(K_1, (\text{ctr}, m_i, i)) & \text{if } \widehat{b} = 0 \\ f_1(\text{ctr}, m_i, i) & \text{if } \widehat{b} = 1 \end{cases}$$

Accordingly, $\mathcal{D}'$ samples $b \xleftarrow{\$} \{0, 1\}$ prepares the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ that on an encryption query on $(\mathbf{M} = m_1 \dots m_n, \widehat{\mathbf{M}} = \widehat{m}_1 \dots \widehat{m}_n) \in \{0, 1\}^n \times \{0, 1\}^n$ from the distinguisher $\overline{\mathcal{D}}$, returns the ciphertext $\text{ct} = (c_1, \dots, c_n) \leftarrow \mathcal{O}_{\text{Enc}}^b(\mathbf{M}, \widehat{\mathbf{M}})$ where

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i \xleftarrow{\$} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = \begin{cases} \mathcal{O}_{\text{PRF}}^b(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = m_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \overline{m}_i = 1 - m_i \end{cases} \quad \text{if } b = 1$$

for $i \in [n]$ and $f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ is a uniformly random function chosen by $\mathcal{D}'$. The distinguisher $\mathcal{D}'$ initializes the counter $\text{ctr} \xleftarrow{\$} \mathbb{Z}_N$ and increases the counter by 1 modulo N if $b = 1$ after each query. The PRF distinguisher $\mathcal{D}'$ invokes $\overline{\mathcal{D}}$ to distinguish between **Game**₁ and **Game**₂. The encryption oracle query made by $\overline{\mathcal{D}}$ to the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ is simulated by the PRF distinguisher $\mathcal{D}'$ using the PRF oracle $\mathcal{O}_{\text{PRF}}^b(\cdot)$ simulated by the PRF challenger $\mathcal{C}'$. Note the following cases.

$$\text{case } b = 1 \quad \widehat{b} = 0 \text{ implies } y_i = \begin{cases} F(K, (\text{ctr}, m_i, i)) & \text{if } m_i = \widehat{m}_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \overline{m}_i = 1 - m_i \end{cases} \text{ as in Game}_1.$$

$$\widehat{b} = 1 \text{ implies } y_i = \begin{cases} f_1(\text{ctr}, m_i, i) & \text{if } m_i = \widehat{m}_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \overline{m}_i = 1 - m_i \end{cases} \text{ as in Game}_2.$$

case $b = 0$ $y_i \xleftarrow{\$} \mathbb{Z}_N^*$ for $\widehat{b} \in \{0, 1\}$ as in both **Game**₁ and **Game**₂.

Thus, the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ simulates **Game**₁ if $\widehat{b} = 0$ and **Game**₂ if $\widehat{b} = 1$.

Now, if the distinguisher $\overline{\mathcal{D}}$ identifies the given game as **Game**₁, $\mathcal{D}'$ outputs $\widehat{b}' = 0$, else $\widehat{b}' = 1$. Whenever $\overline{\mathcal{D}}$ succeeds in distinguishing **Game**₁ and **Game**₂, $\mathcal{D}'$ wins the experiment $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$. Therefore, $\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] \geq \frac{1}{2} + \delta$ since we assumed that $\overline{\mathcal{D}}$ distinguishes **Game**₁ and **Game**₂ with probability $\frac{1}{2} + \delta$. Consequently,

$$\text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} = |\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] - \frac{1}{2}| \geq |\frac{1}{2} + \delta - \frac{1}{2}| = \delta$$

Since F is a secure PRF (Def. 14),

$$\delta \leq \text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} \leq \epsilon_2(\lambda)$$

for some negligible function $\epsilon_2(\cdot)$. Thus any distinguisher $\overline{\mathcal{D}}$ can distinguish between **Game**₁ and **Game**₂ only with probability negligibly close to $\frac{1}{2}$. Hence,

$$|\Pr[E_1] - \Pr[E_2]| < \epsilon_2(\lambda)$$

□

- **Game**₃: This is a modified game where the distinguisher $\mathcal{D}$, on encryption oracle query $(\mathbf{M} = m_1 \dots m_n, \widehat{\mathbf{M}} = \widehat{m}_1 \dots \widehat{m}_n) \in \{0, 1\}^n \times \{0, 1\}^n$, receives the ciphertext $\text{ct} = (c_1, \dots, c_n)$ where

$$c_i = y_i^2 x^{m_i} \bmod N$$

with

$$y_i \xleftarrow{\$} \mathbb{Z}_N^* \quad \text{if } b = 0$$

$$y_i = f(\text{ctr}, m_i, i) \quad \text{if } b = 1$$

for all $i \in [n]$ and $f : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ is a uniformly random function. Thus, this game replaces the PRF F used in Game_0 for simulating the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ by a uniformly random function $f : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$.

The only difference between Game_2 and Game_3 is that when $b = 1$,

$$y_i = \begin{cases} f_1(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = m_i \\ f_2(\text{ctr}, m_i, i) & \text{if } \widehat{m}_i = \overline{m}_i = 1 - m_i \end{cases}$$

in Game_2 whereas in Game_3 , $y_i = f(\text{ctr}, m_i, i)$ for all $i \in [n]$ where $f, f_1, f_2 : \mathbb{Z}_N \times \{0, 1\} \times [n] \rightarrow \mathbb{Z}_N^*$ are uniformly random functions. Consequently, Game_2 is indistinguishable from Game_3 . Hence, $\Pr[E_2] = \Pr[E_3]$.

In Game_3 , for $b = 0$, y_i is uniformly random element from $\mathbb{Z}_N^*$. Since f is uniformly random function and the input ctr is updated by the challenger for each encryption oracle query for $b = 1$, the output of f remains random during multiple queries even on the same bit m_i . Thus y_i remains effectively uniformly random during multiple queries for $b = 1$. Since in a single query, for different bits, the third element i (the position of the bit) in the input tuple of f gets changed, the output y_i of f remains random. Thus, $\Pr[E_3] = \frac{1}{2}$.

Therefore,

$$\begin{aligned} \text{Adv}_{\Pi_{\text{aGM}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) &= \left| \Pr \left[\text{Exp}_{\Pi_{\text{aGM}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) = 1 \right] - \frac{1}{2} \right| = \left| \Pr[E_0] - \frac{1}{2} \right| \\ &\leq |\Pr[E_0] - \Pr[E_1]| + |\Pr[E_1] - \Pr[E_2]| + |\Pr[E_2] - \Pr[E_3]| + \left| \Pr[E_3] - \frac{1}{2} \right| \\ &< \epsilon_1(\lambda) + \epsilon_2(\lambda) + 0 + 0 \\ &= \text{neg}(\lambda) \end{aligned}$$

Hence, the AE Π_{aGM} of Goldwasser-Micali cryptosystem Π_{GM} is IND-NA secure. $\square$

4 Our construction Π_{aBen} : synchronous AE from Benaloh PKE

Let Π_{Ben} be the Benaloh PKE discussed in Section 2.2 associated with the plaintext space $\mathcal{M} = \mathbb{Z}_r$ and the ciphertext space $\mathcal{C} = \mathbb{Z}_N$ where r is a prime and $N = pq$ for primes p, q satisfying $r \mid (p-1), r^2 \nmid (p-1), r \nmid (q-1)$. We describe below its AE Π_{aBen} associated with the covert plaintext space $\widehat{\mathcal{M}} = \Delta = \{a^r \mid a \in \mathbb{Z}_N^*\}$. We use the following notation.

- $\Delta = \{a^r \mid a \in \mathbb{Z}_N^*\}$. Note that $|\Delta| = \frac{\Phi(N)}{r}$.
- $\mathcal{F} = \{F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta \mid F \text{ is a PRF}\}$.

When $\Pi_{\text{Ben}} = \begin{bmatrix} \text{Setup}(\lambda) \rightarrow \text{pp} \\ \text{Gen}(\text{pp}) \rightarrow (\text{sk}, \text{pk}) \\ \text{Enc}(\text{pp}, \text{M}, \text{pk}) \rightarrow \text{ct} \\ \text{Dec}(\text{pp}, \text{ct}, \text{sk}) \rightarrow \text{M} \end{bmatrix}$ is in active mode, the normal public parameter

$\text{pp} = \mathcal{G}_\lambda$ is generated using the $\Pi_{\text{Ben}}.\text{Setup}$ algorithm where $\mathcal{G}_\lambda$ generates pair of random λ -bit primes p and q such that $r \mid (p-1), r^2 \nmid (p-1), r \nmid (q-1)$. Each user runs the algorithm $\Pi_{\text{Ben}}.\text{Gen}$ on input pp and generates its corresponding secret key $\text{sk} = (\phi, x)$ and public key $\text{pk} = (y, N)$ where $N = pq$ for the λ -bit prime-pair (p, q)

generated by $\mathcal{G}_\lambda$, $\phi = \Phi(N) = (p-1)(q-1)$, $y \in \mathbb{Z}_N^*$ and $x = y^{\phi/r} \bmod N$ such that $x \neq 1 \bmod N$. The secret key is kept secret to the user itself whereas the public key is

published by the user. When $\Pi_{\text{aBen}} = \left[\begin{array}{l} \text{Setup}(\text{pk}) \rightarrow \text{ap} \\ \text{Gen}(\text{ap}, \text{pk}) \rightarrow (\text{dk}, \text{ctr}) \\ \text{Enc}(\text{ap}, \text{M}, \widehat{\text{M}}, \text{dk}, \text{ctr}) \rightarrow (\text{ct}, \text{ctr}) \\ \text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \rightarrow (\widehat{\Gamma}, \text{ctr}) \end{array} \right]$ is in active mode, the four PPT algorithms of Π_{aBen} are executed.

- The algorithm $\Pi_{\text{aBen}}.\text{Setup}$ presented in Algorithm 14 is run once on input the existing public key $\text{pk} = (y, N)$ of the receiver at the beginning to set the anamorphic public parameters ap which contains a randomly chosen PRF $F \in \mathcal{F}$ and ap is published publicly.

Algorithm 14 $\Pi_{\text{aBen}}.\text{Setup}(\text{pk}) \rightarrow \text{ap}$

Require: the public key $\text{pk} = (y, N)$ of the receiver where $N = pq$ for two large λ -bit primes p, q such that $r \mid (p-1)$, $r^2 \nmid (p-1)$, $r \nmid (q-1)$ where the prime r is plaintext space size, $y \in \mathbb{Z}_N^*$.

Ensure: anamorphic public parameter ap .

- 1: $F \xleftarrow{\$} \mathcal{F}$ where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is an arbitrarily chosen PRF from $\mathcal{F}$.
 - 2: $\text{ap} = F$;
-

- The sender (resp. the receiver) runs the algorithm $\Pi_{\text{aBen}}.\text{Gen}$ given in Algorithm 15 on input the anamorphic public parameter $\text{ap} = F$ and the public key $\text{pk} = (y, N)$ of the receiver (secret key holder), generates the double key $\text{dk} = (K, \text{pk})$ and a counter $\text{ctr} \in \mathbb{Z}_N$ and communicates (dk, ctr) to the receiver (resp. the sender) through a secure channel where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a PRF, $N = pq$ for the λ -bit primes p, q , $y \in \mathbb{Z}_N^*$ and ctr is initialized with an integer modulo N .

Algorithm 15 $\Pi_{\text{aBen}}.\text{Gen}(\text{ap}, \text{pk}) \rightarrow (\text{dk}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$, public key $\text{pk} = (y, N)$.

Ensure: double key dk and counter ctr .

- 1: $K \xleftarrow{\$} \mathbb{Z}_N$;
 - 2: $\text{ctr} \xleftarrow{\$} \mathbb{Z}_N$;
 - 3: $\text{dk} = (K, \text{pk} = (y, N))$;
-

- The algorithm $\Pi_{\text{aBen}}.\text{Enc}$ discussed in Algorithm 16 is executed by the sender on input the anamorphic public parameters $\text{ap} = F$, a normal plaintext $\text{M} \in \mathcal{M}$, a covert plaintext $\widehat{\text{M}} \in \widehat{\mathcal{M}}$, the double key $\text{dk} = (K, \text{pk} = (y, N))$ of the receiver and the current counter $\text{ctr} \in \mathbb{Z}_N$ and outputs a ciphertext $\text{ct} \in \mathcal{C}$ and an updated counter $\text{ctr} \in \mathbb{Z}_N$ where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a PRF, $N = pq$ for the λ -bit primes p, q , $y \in \mathbb{Z}_N^*$, $K \in \mathbb{Z}_N$.

Algorithm 16 $\Pi_{\text{aBen}}.\text{Enc}(\text{ap}, \text{M}, \widehat{\text{M}}, \text{dk}, \text{ctr}) \rightarrow (\text{ct}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$, normal plaintext $\text{M} \in \mathcal{M} = \mathbb{Z}_r$, anamorphic plaintext $\widehat{\text{M}} \in \widehat{\mathcal{M}} = \Delta$, double key $\text{dk} = (K, \text{pk} = (y, N))$ and current counter $\text{ctr} \in \mathbb{Z}_N$.

Ensure: ciphertext $\text{ct} \in \mathcal{C} = \mathbb{Z}_N$.

- 1: $\widehat{u} = \widehat{\text{M}} \times F(K, (\text{ctr}, \text{M})) \bmod N \in \Delta$;
 - 2: $\text{ct} = y^{\text{M}} \widehat{u} \bmod N \in \mathbb{Z}_N$;
 - 3: $\text{ctr} = \text{ctr} + 1 \bmod N$;
-

- The algorithm $\Pi_{\text{aBen}}.\text{Dec}$ described in Algorithm 17 is run by the receiver on input the anamorphic parameter $\text{ap} = F$, the ciphertext $\text{ct} \in \mathcal{C} = \mathbb{Z}_N$ and its own secret key $\text{sk} = (\phi, x)$, double key $\text{dk} = (K, \text{pk} = (y, N))$ and the current counter $\text{ctr} \in \mathbb{Z}_N$ to recover a covert plaintext $\widehat{\Gamma} \in \widehat{\mathcal{M}}$ and an updated counter ctr where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a PRF, $N = pq$ for the λ -bit primes p, q , $\phi = \Phi(N) = (p-1)(q-1)$, $y \in \mathbb{Z}_N^*$ such that $y^{\phi/r} \not\equiv 1 \pmod{N}$, $x \equiv y^{\phi/r} \pmod{N} \in \mathbb{Z}_N$ and $K \in \mathbb{Z}_N$.

Algorithm 17 $\Pi_{\text{aBen}}.\text{Dec}(\text{ap}, \text{ct}, \text{sk}, \text{dk}, \text{ctr}) \rightarrow (\widehat{\Gamma}, \text{ctr})$

Require: anamorphic public parameter $\text{ap} = F$, ciphertext $\text{ct} \in \mathcal{C} = \mathbb{Z}_N$, secret key $\text{sk} = (\phi, x)$, double key $\text{dk} = (K, \text{pk} = (y, N))$ and current counter $\text{ctr} \in \mathbb{Z}_N$.

Ensure: plaintext $\widehat{\Gamma} \in \widehat{\mathcal{M}} = \Delta$ and updated counter ctr .

- 1: $\text{M} = \perp$;
 - 2: $\widehat{\Gamma} = \perp$;
 - 3: $a = \text{ct}^{\phi/r} \bmod N$;
 - 4: **for** ($i = 0$ to $r - 1$) **do**
 - 5: **if** ($x^i \equiv a \pmod{N}$) **then**
 - 6: $\text{M} = i$;
 - 7: **break**;
 - 8: **end if**
 - 9: **end for**
 - 10: $u' = \text{ct}(y^{-1})^{\text{M}} \bmod N$ where y^{-1} is the multiplicative inverse of $y \in \mathbb{Z}_N^*$;
 - 11: $\widehat{\Gamma} = u' \times F(K, (\text{ctr}, \text{M}))^{-1}$ where $F(K, (\text{ctr}, \text{M}))^{-1}$ is the multiplicative inverse of $F(K, (\text{ctr}, \text{M})) \in \Delta \subseteq \mathbb{Z}_N^*$;
 - 12: $\text{ctr} = \text{ctr} + 1 \bmod N$;
-

Theorem 4.1 *Our proposed AE $\Pi_{\text{aBen}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ (presented in Section 4) of Benaloh cryptosystem $\Pi_{\text{Ben}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is correct (as per Def. 6).*

Proof Let

$$\begin{aligned}
 \text{pp} &= \mathcal{G}_\lambda \leftarrow \Pi_{\text{BG}}.\text{Setup}(1^\lambda) \\
 (\text{sk} = (\phi, x), \text{pk} = (y, N)) &\leftarrow \Pi_{\text{BG}}.\text{Gen}(\text{pp}) \\
 \text{ap} &= F \leftarrow \Pi_{\text{aBen}}.\text{Setup}(\text{pk}) \\
 (\text{dk} = (K, \text{pk} = (y, N)), \text{ctr}) &\leftarrow \Pi_{\text{aBen}}.\text{Gen}(\text{ap}, \text{pk})
 \end{aligned}$$

where λ is the security parameter, $\mathcal{G}_\lambda$ is large λ -bit prime-pair generator that generates prime numbers p, q such that $r \mid (p-1), r^2 \nmid (p-1), r \nmid (q-1)$, r (prime) is the plaintext space

size, $\phi = \Phi(N) = (p-1)(q-1)$, $y \in \mathbb{Z}_N^*$ such that $y^{\phi/r} \neq 1 \pmod{N}$ and $x = y^{\phi/r} \pmod{N}$, $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a randomly chosen PRF, $K \in \mathbb{Z}_N$ and ctr is the initial counter.

Note that ct generated by $\Pi_{\text{aBen}}.\text{Enc}$ (see lines 1 – 2 of Algorithm 16) satisfies

$$\text{ct} = y^{\text{M}} \hat{u} \pmod{N} = y^{\text{M}} \hat{\text{M}} \times F(K, (\text{ctr}, \text{M})) \pmod{N} \quad (3)$$

as $\hat{u} = \hat{\text{M}} \times F(K, (\text{ctr}, \text{M})) \pmod{N}$.

Note that $y \in \mathbb{Z}_N^*$ implies y^{-1} exists and $F(K, (\text{ctr}, \text{M})) \in \Delta \subseteq \mathbb{Z}_N^*$ implies $F(K, (\text{ctr}, \text{M}))^{-1}$ exists as $\mathbb{Z}_N^*$ is a multiplicative group.

The normal plaintext M is recovered in the algorithm $\Pi_{\text{aBen}}.\text{Dec}$ (lines 3 – 9 of Algorithm 17) in the same way as in $\Pi_{\text{Ben}}.\text{Dec}$ (lines 2 – 8 of Algorithm 9) and is therefore correct by the correctness of Π_{Ben} as shown in Section 2.2. Now from lines 11 – 12 of Algorithm 17

$$\begin{aligned} \hat{\Gamma} &= u' \times \left(F(K, (\text{ctr}, \text{M})) \right)^{-1} \pmod{N} \\ &= \text{ct}(y^{-1})^{\text{M}} \times \left(F(K, (\text{ctr}, \text{M})) \right)^{-1} \pmod{N} \quad [\text{as } u' = \text{ct}(y^{-1})^{\text{M}} \pmod{N}] \\ &= y^{\text{M}} \hat{\text{M}} \times F(K, (\text{ctr}, \text{M})) \times y^{-\text{M}} \times \left(F(K, (\text{ctr}, \text{M})) \right)^{-1} \pmod{N} \quad [\text{by Eq. 3}] \\ &= \hat{\text{M}} \pmod{N} \end{aligned}$$

Consequently, $\Pr[\hat{\Gamma} = \hat{\text{M}}] = 1$ and the correctness of Π_{aBen} follows. $\square$

Theorem 4.2 *The AE $\Pi_{\text{aBen}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ (presented above) of Benaloh cryptosystem $\Pi_{\text{Ben}} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec})$ is IND-NA secure as per the security model described in Def. 8 assuming the existence of a PRF as per Def. 14.*

Proof We describe below the security game $\text{Game}_0 = \text{Exp}_{\text{aBen}, \mathcal{D}}^{\text{IND-NA}}$ between a challenger $\mathcal{C}$ and a distinguisher $\mathcal{D}$ following Fig. 2.

Setup phase: The challenger generates the public parameter $\text{pp} = \mathcal{G}_\lambda \leftarrow \Pi_{\text{Ben}}.\text{Setup}(1^\lambda)$, a secret key and public key pair $(\text{sk} = (\phi, x), \text{pk} = (y, N)) \leftarrow \Pi_{\text{Ben}}.\text{Gen}(\text{pp})$ and sends $(\text{pp}, \text{sk}, \text{pk})$ to the distinguisher $\mathcal{D}$. where $\mathcal{G}_\lambda$ generates λ -bit prime-pairs p, q such that $r \mid (p-1), r^2 \nmid (p-1), r \nmid (q-1)$ for a prime number r being the plaintext space size, (p, q) is a prime-pair generated by $\mathcal{G}_\lambda$, $N = pq$, $\phi = \Phi(N) = (p-1)(q-1)$, $y \in \mathbb{Z}_N^*$ such that $y^{\phi/r} \neq 1 \pmod{N}$ and $x = y^{\phi/r} \pmod{N}$.

The challenger $\mathcal{C}$ also generates the anamorphic public parameter $\text{ap} = F \leftarrow \Pi_{\text{aBen}}.\text{Setup}(\text{pk})$ and the double key-counter pair $(\text{dk} = (K, \text{pk}), \text{ctr}) \leftarrow \Pi_{\text{aBen}}.\text{Gen}(\text{ap}, \text{pk})$ where $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a randomly chosen PRF from the family of PRFs $\mathcal{F}$, $K \in \mathbb{Z}_N$, $\text{ctr} \in \mathbb{Z}_N$ and ap is sent to $\mathcal{D}$.

Challenge phase: The challenger $\mathcal{C}$ samples a bit $b \xleftarrow{\$} \{0, 1\}$ and gives the distinguisher $\mathcal{D}$ access to the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ where

$$\mathcal{O}_{\text{Enc}}^b(\text{M}, \hat{\text{M}}) = \text{ct}$$

with

$$\begin{aligned} \text{ct} &\leftarrow \Pi_{\text{Ben}}.\text{Enc}(\text{pp}, \text{M}, \text{pk}) && \text{if } b = 0 \\ (\text{ct}, \text{ctr}) &\leftarrow \Pi_{\text{aBen}}.\text{Enc}(\text{ap}, \text{M}, \hat{\text{M}}, \text{dk}, \text{ctr}) && \text{if } b = 1 \end{aligned}$$

for $(\text{M}, \hat{\text{M}}) \in \mathcal{M} \times \hat{\mathcal{M}} = \mathbb{Z}_r \times \Delta$ and $\text{ctr} \in \mathbb{Z}_N$ is the current counter which stays unchanged and unutilized if $b = 0$ and gets updated by the challenger $\mathcal{C}$ after each oracle query to $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ if $b = 1$ as per Line 3 of Algorithm 16. Note that

$$\text{ct} = y^{\text{M}} \hat{u} \pmod{N}$$

with

$$\begin{aligned} \hat{u} &= u^r, u \xleftarrow{\$} \mathbb{Z}_N^* & \text{if } b = 0 \\ \hat{u} &= \hat{M}u' \bmod N, u' = F(K, (\text{ctr}, M)) & \text{if } b = 1 \end{aligned}$$

Query phase: The distinguisher $\mathcal{D}$ can query polynomially many times to the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ on pairs of normal and covert plaintexts $(M, \hat{M}) \in \mathcal{M} \times \hat{\mathcal{M}}$.

Guess phase: Finally, the distinguisher $\mathcal{D}$ outputs a bit $b' \in \{0, 1\}$ as a guess for the bit b to challenger and wins the game if $b = b'$.

We define three more games **Game**₁, **Game**₂ and **Game**₃. Let E_i is the event that a distinguisher wins the game **Game** _{i} for $i \in [3]$. The distinguisher $\mathcal{D}$ makes query of the form $(M, \hat{M}) \in \mathcal{M} \times \hat{\mathcal{M}} = \mathbb{Z}_r \times \Delta$ to the challenger $\mathcal{C}$ in different games as shown below and receives different response ciphertext $\text{ct} \in \mathbb{Z}_N$ accordingly.

- **Game**₁: In this game, the distinguisher $\mathcal{D}$, on encryption oracle query $(M, \hat{M}) \in \mathbb{Z}_r \times \Delta$ receives the ciphertext $\text{ct} \in \mathbb{Z}_N$ where

$$\text{ct} = y^M \hat{u} \bmod N$$

with

$$\begin{aligned} \hat{u} &= u^r, u \xleftarrow{\$} \mathbb{Z}_N^* & \text{if } b = 0 \\ \hat{u} &= \hat{M}u' \bmod N, u' = f(\text{ctr}, M) & \text{if } b = 1 \end{aligned}$$

for an uniformly random function $f : \mathbb{Z}_N \times \mathbb{Z}_r \rightarrow \Delta$.

Claim 3 **Game**₀ is indistinguishable from **Game**₁.

Proof The **Game**₀ and **Game**₁ are identical except that for $b = 1$, $u' = F(K, (\text{ctr}, M))$ in **Game**₀, whereas in **Game**₁, $u' = f(\text{ctr}, M)$ where $f : \mathbb{Z}_N \times \mathbb{Z}_r \rightarrow \Delta$ is a uniformly random function and $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ is a secure PRF. Suppose a distinguisher $\overline{\mathcal{D}}$ distinguishes between **Game**₀ and **Game**₁ with probability $\frac{1}{2} + \delta$ for some $\delta \in \mathbb{R}$ such that $0 \leq \delta \leq \frac{1}{2}$.

We construct below a PRF distinguisher $\mathcal{D}'$ for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ in the PRF security game $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$ (see Fig. 3) against PRF challenger $\mathcal{C}'$ and show how $\mathcal{D}'$ invokes $\overline{\mathcal{D}}$.

Firstly, $\mathcal{C}'$ samples $K \xleftarrow{\$} \mathcal{K} = \mathbb{Z}_N$ and a uniformly random function $f : \mathbb{Z}_N \times \mathbb{Z}_r \rightarrow \Delta$. Then the PRF challenger $\mathcal{C}'$ samples a bit $\hat{b} \xleftarrow{\$} \{0, 1\}$, and gives $\mathcal{D}'$ the access to the PRF oracle $\mathcal{O}_{\text{PRF}}^{\hat{b}}(\cdot)$ where $\mathcal{O}_{\text{PRF}}^0(\cdot)$ is the oracle for the PRF $F : \mathbb{Z}_N \times (\mathbb{Z}_N \times \mathbb{Z}_r) \rightarrow \Delta$ and $\mathcal{O}_{\text{PRF}}^1(\cdot)$ is the oracle for the uniformly random function $f : \mathbb{Z}_N \times \mathbb{Z}_r \rightarrow \Delta$. Thus,

$$\mathcal{O}_{\text{PRF}}^{\hat{b}}(\text{ctr}, M) = \begin{cases} F(K, (\text{ctr}, M)) & \text{if } \hat{b} = 0 \\ f(\text{ctr}, M) & \text{if } \hat{b} = 1 \end{cases}$$

Accordingly, $\mathcal{D}'$ samples a bit $b \xleftarrow{\$} \{0, 1\}$ and prepares the oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ that on an encryption query on $(M, \hat{M}) \in \mathbb{Z}_r \times \Delta$ from the distinguisher $\overline{\mathcal{D}}$ returns the ciphertext $\text{ct} \leftarrow \mathcal{O}_{\text{Enc}}^b(M, \hat{M})$ where

$$\text{ct} = y^M \hat{u} \bmod N$$

with

$$\begin{aligned} \hat{u} &= u^r, u \xleftarrow{\$} \mathbb{Z}_N^* & \text{if } b = 0 \\ \hat{u} &= \hat{M}u' \bmod N, u' = \mathcal{O}_{\text{PRF}}^{\hat{b}}(\text{ctr}, M) & \text{if } b = 1 \end{aligned}$$

The distinguisher $\mathcal{D}'$ initializes the counter $\text{ctr} \xleftarrow{\$} \mathbb{Z}_N$ and increases the counter by 1 modulo N if $b = 1$ after each query. The PRF distinguisher $\mathcal{D}'$ invokes $\overline{\mathcal{D}}$ to distinguish between Game_0 and Game_1 . Note the following cases.

case $b = 1$ $\hat{b} = 0$ implies $\hat{u} = \hat{M}u' \bmod N$, $u' = F(K, (\text{ctr}, M))$

$\hat{b} = 1$ implies $\hat{u} = \hat{M}u' \bmod N$, $u' = f(\text{ctr}, M)$

case $b = 0$ $\hat{u} = u^r$, $u \xleftarrow{\$} \mathbb{Z}_N^*$ for $\hat{b} \in \{0, 1\}$

Thus, the encryption oracle $\mathcal{O}_{\text{Enc}}^b(\cdot, \cdot)$ simulates Game_0 if $\hat{b} = 0$ and Game_1 if $\hat{b} = 1$.

Now, if the distinguisher $\overline{\mathcal{D}}$ identifies the given game as Game_0 , $\mathcal{D}'$ outputs $\hat{b}' = 0$, else $\hat{b}' = 1$. Whenever $\overline{\mathcal{D}}$ succeeds in distinguishing between Game_0 and Game_1 , $\mathcal{D}'$ wins the experiment $\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}$. Therefore, $\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] \geq \frac{1}{2} + \delta$ since we assumed that $\overline{\mathcal{D}}$ distinguishes between Game_0 and Game_1 with probability $\frac{1}{2} + \delta$. Consequently,

$$\text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} = |\Pr[\text{Exp}_{F, \mathcal{D}'}^{\text{IND-PRF}}(\lambda) = 1] - \frac{1}{2}| \geq |\frac{1}{2} + \delta - \frac{1}{2}| = \delta$$

Since F is a secure PRF (Def. 14),

$$\delta \leq \text{Adv}_{F, \mathcal{D}'}^{\text{IND-PRF}} \leq \epsilon(\lambda)$$

for some negligible function $\epsilon(\cdot)$. Thus any distinguisher $\overline{\mathcal{D}}$ can distinguish between Game_0 and Game_1 only with probability negligibly close to $\frac{1}{2}$. Hence,

$$|\Pr[E_0] - \Pr[E_1]| < \epsilon(\lambda)$$

□

- **Game₂**: In this game, the distinguisher $\mathcal{D}$, on encryption oracle query $(M, \hat{M}) \in \mathbb{Z}_r \times \Delta$, receives the ciphertext $\text{ct} \in \mathbb{Z}_N$ where

$$\text{ct} = y^M \hat{u} \bmod N$$

with

$$\begin{aligned} \hat{u} &= u^r, u \xleftarrow{\$} \mathbb{Z}_N^* & \text{if } b = 0 \\ \hat{u} &= \hat{M}u' \bmod N, u' \xleftarrow{\$} \Delta & \text{if } b = 1 \end{aligned}$$

The only difference in Game_1 and Game_2 is that in Game_1 , $u' = f(\text{ctr}, M)$ whereas u' is randomly chosen from Δ in Game_2 for $b = 1$ where $f : \mathbb{Z}_N \times \mathbb{Z}_r \rightarrow \Delta$ is a uniformly random function. Since f is a uniformly random function and ctr gets updated after each encryption query by the challenger $\mathcal{C}$, the output of f remains effectively random from the space Δ . Thus, the winning probability of $\mathcal{D}$ in Game_1 is $\Pr[E_1] = \Pr[E_2] = \frac{1}{2}$.

- **Game₃**: This is a modified game where the distinguisher $\mathcal{D}$, on encryption oracle query $(M, \hat{M}) \in \mathbb{Z}_r \times \Delta$, receives the ciphertext $\text{ct} \in \mathbb{Z}_N$ where

$$\text{ct} = y^M \hat{u} \bmod N$$

with

$$\begin{aligned} \hat{u} &= u^r, u \xleftarrow{\$} \mathbb{Z}_N^* & \text{if } b = 0 \\ \hat{u} &\xleftarrow{\$} \Delta & \text{if } b = 1 \end{aligned}$$

Since $\widehat{M} \in \Delta$ and u' is randomly chosen from Δ , $\widehat{u}$ is effectively a random element from Δ in Game_2 similar to Game_3 . Hence, the winning probability of $\mathcal{D}$ in Game_2 is $\Pr[E_2] = \Pr[E_3] = \frac{1}{2}$.

Since $\Delta = \{x^r \mid x \in \mathbb{Z}_N^*\}$, $\Pr[E_3] = \frac{1}{2}$.

Therefore,

$$\begin{aligned} \text{Adv}_{\Pi_{\text{aBen}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) &= \left| \Pr \left[\text{Exp}_{\Pi_{\text{aBen}}, \mathcal{D}}^{\text{IND-NA}}(\lambda) = 1 \right] - \frac{1}{2} \right| = \left| \Pr[E_0] - \frac{1}{2} \right| \\ &\leq |\Pr[E_0] - \Pr[E_1]| + |\Pr[E_1] - \Pr[E_2]| + |\Pr[E_2] - \Pr[E_3]| + \left| \Pr[E_3] - \frac{1}{2} \right| \\ &< \epsilon(\lambda) + 0 + 0 + 0 = \epsilon(\lambda) \end{aligned}$$

Hence, the AE Π_{aBen} of Benaloh cryptosystem Π_{Ben} is IND-NA secure. $\square$

5 Conclusion

In this paper, we have proposed two concrete AE schemes from the Goldwasser-Micali PKE and Benaloh PKE. We have proved them to be IND-NA secure from the existence of secure PRF. Finally, our first construction supports natural robustness. Both the proposed schemes provide high bandwidth rate, low anamorphic ciphertext expansion rate, small double key size, low computational cost and strong deniability. More significantly, we have utilized the underlying structures of the PKEs to enable unique and efficient AEs.

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